

Attention Scarcity and Optimal Depth in AI Delegation Chains

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Abstract

When AI agents delegate to other AI agents, the scarcity of attention—not the cost of coordination—bounds organizational depth. We embed rational inattention into a multi-layer hierarchy: each agent allocates scarce attention across upstream signals subject to a finite budget (context window). The endogenous transmission factor emerges from Bayesian updating rather than being assumed. We prove that optimal depth is finite and interior, that precision loss compounds geometrically through the chain, and that front-loaded budget allocation—larger context windows at early layers—maximizes output quality. When API costs vanish, the bound on depth persists because finite context windows ensure that each layer must compress information, and that compression compounds. We calibrate the model to LLM benchmark data and propose five testable predictions with specific regression specifications and identification strategies.

JEL Codes: D23, D82, L22, L86, O32, O33

1 Introduction

Enterprise AI systems increasingly rely on multi-agent architectures, where complex tasks are decomposed across chains of specialized language models. A growing empirical literature documents the performance of these systems—yet the fundamental trade-off governing their optimal depth remains theoretically unexplored. As context windows expand by orders of magnitude (from GPT-3’s 2K tokens to Gemini-1.5’s 1M tokens) and API costs continue to decline, the natural intuition is that deeper chains—more layers of specialized processing—should become optimal. We show that this intuition is incomplete: even with costless communication, endogenous constraints on information transmission bound the efficient depth of the chain.

The core economic mechanism is *information depreciation*—the gradual loss of signal quality as information traverses a processing hierarchy. In a multi-agent chain, each layer receives noisy signals from the layer below, updates beliefs, and transmits a summary upward. Because each agent’s context window is finite, not all incoming signals can be attended to with full precision; some are compressed, some are discarded, and the resulting transmission is necessarily coarser than the input. This degradation compounds

across layers, so that the signal reaching the terminal layer carries only a fraction of the original information content. Information depreciation thus operates analogously to capital depreciation: each processing layer is an investment in refined decision-making, but that investment suffers a depreciative loss before it reaches the next stage.

We formalize this intuition by embedding rational inattention (Sims 2003) into a multi-layer information-processing hierarchy. Two conditions jointly generate finite optimal depth. First, signals depreciate in quality as they traverse the chain, formalized as an exogenous retention factor $\eta \in (0, 1)$ that captures the natural decay of information during processing (Proposition 8). Second, each agent faces a finite Shannon-capacity budget for attention allocation, which determines *how much* of the depreciated signal survives at each layer. The depth of the chain is bounded because the compounding of these two forces—exogenous depreciation and endogenous attention loss—eventually renders marginal layers unproductive. Either condition alone is insufficient: with $\eta = 1$ and no exogenous degradation, information would circulate without bound; without a binding attention constraint, the retention rate could be maintained regardless of depth.

Our analysis is related to the canonical models of multi-layer organization developed by Garicano (2000) and Dessein and Santos (2006) (henceforth, GHM). In the GHM framework, finite organizational depth arises from the interplay of diminishing returns and increasing marginal costs of communication or effort. While information loss in hierarchies is a familiar theme in organizational economics (e.g., ???), existing models typically treat communication fidelity as an exogenous parameter or as the outcome of explicit incentive problems. Our model *complements* the GHM framework by adding a micro-foundation for information processing: the transmission factor ρ_ℓ , defined as the ratio of achieved posterior precision to first-best posterior precision, emerges endogenously from the optimal attention-allocation problem and reflects the actual information degradation at each layer. This connects the organizational model to the Bayesian micro-foundations of rational inattention (Sims 2003) and generates predictions about how attention weights, signal heterogeneity, and capacity constraints jointly determine organizational form.

This paper makes three contributions. First, we introduce the concept of *information depreciation* into the theory of organizational hierarchies. By modeling signal quality as a depreciating asset that traverses a processing chain, we provide a tractable framework that links communication-loss models in organizational economics (??) to the rational-inattention literature (Sims 2003). The depreciation parameter η discipline the model’s predictions and can be calibrated from observable features of LLM systems.

Second, we provide a mapping from LLM architecture parameters—context window length, token-level precision, and API pricing—to predictions about optimal organizational depth. The model generates sharp comparative statics: depth increases with context window size and decreases with signal depreciation; API costs affect the intensive margin of processing but are largely irrelevant for the extensive margin of depth once costs are below a threshold. These predictions are amenable to empirical testing and can guide system-design decisions in enterprise AI deployment.

Third, we calibrate the model using publicly available data on benchmark accuracy, context window capacities, and API pricing for state-of-the-art language models. The calibration demonstrates that the key predictions hold for empirically plausible parameter values and generates quantitative predictions about the range of efficient depth under current and projected technology. We provide comparative statics that can discipline future empirical work on multi-agent AI architectures.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model. Section 4 analyzes the single-layer attention-allocation problem and derives the endogenous transmission factor ρ_ℓ . Section 5 extends the analysis to multiple layers and characterizes optimal depth. Section 6 calibrates the model using LLM technology data and presents comparative statics. Section 7 discusses testable implications for the design of multi-agent systems. Section 8 considers extensions, and Section 9 concludes.

2 Related Literature

Our work bridges four literatures: organizational economics, rational inattention, AI and organizations, and information design.

2.1 Organizations and Knowledge Hierarchies

The organizational economics literature on knowledge hierarchies and multi-layer contracting studies how information loss and control loss bound depth. Garicano (2000) provides the canonical model of knowledge-based hierarchies, where problems are routed to agents with appropriate expertise. Melumad, Mookherjee, and Reichelstein (1995) analyze how control loss in hierarchies creates a trade-off between specialization and coordination. Lash and Wellington (2008) explicitly models control loss as a constraint on depth. Bolton and Dewatripont (1994) study the firm as a communication network. Radner (1993) analyze decentralized information processing. Our contribution is to microfound the distortion mechanism in attention allocation rather than assuming it.

2.2 Rational Inattention

The rational inattention literature studies how information-processing constraints shape decision-making. Sims (2003) introduces the rational inattention framework, where agents face an entropy constraint on information processing. Sims (2006) applies this to monetary policy. Mackowiak, Matějka, and Wiederholt (2023) provides a comprehensive review. Dessein and Santos (2006) study organizational attention in a two-layer setting, showing how attention scarcity shapes delegation. Dessein, Galeotti, and Santos (2016) extend this to communication in organizations. Caplin and Dean (2015) develop revealed-preference tests for rational inattention. Our model extends the organizational attention framework to arbitrarily deep chains.

2.3 Multi-Stage Information Design

A growing literature studies information design in dynamic and multi-stage settings. In these environments, a sender strategically controls the flow of information to a receiver (or receivers) who take actions over time. Our paper shares with this literature the fundamental premise that information is transmitted and evolves across multiple stages, but differs in emphasizing constrained information *processing*—arising from attention scarcity within hierarchical organizations—rather than strategic information *withholding* by a sender.

Ely (2017) studies a dynamic information transmission problem in which a sender communicates the state of a Markov process to a receiver via binary signals (“beeps”).

The optimal signaling strategy involves selective disclosure: the sender pools states and releases information only when it is most consequential for the receiver’s decision. The key insight is that the *timing* of information revelation is a strategic choice, and the sender may optimally delay or suppress information to influence the receiver’s beliefs. Our framework differs in that the constraint on information transmission arises not from a sender’s strategic objectives but from the organizational limitation that each processing layer has finite attention capacity.

Renault, Solan, and Vieille (2017) analyze optimal dynamic information provision in a sender-receiver framework with a persistent binary state. They characterize the sender’s value function and show that the optimal policy involves gradual revelation: information is released piecemeal over time to maximize the sender’s expected payoff. The dynamics of belief updating are central to their analysis, as the sender faces a trade-off between revealing information early and withholding it to preserve future influence. In our model, similar intertemporal trade-offs arise, but they are driven by organizational rather than strategic considerations.

Escudé and Sinander (2023) study “slow persuasion,” in which a sender provides public information over time subject to a graduality constraint. They show that when information can only accumulate gradually, the sender’s ability to persuade is inhibited relative to the unconstrained benchmark. The graduality constraint in their model is exogenous, much as the attention constraint in our framework is determined by the organizational technology. The key difference is that their constraint limits the *speed* of information acquisition, whereas ours limits the *volume* of information that can be processed at each layer.

Smolin (2021) studies a principal who designs a dynamic evaluation policy for an agent of uncertain productivity. Smolin shows that the equilibrium is Pareto efficient and that the wage deterministically grows with tenure. The evaluation policy serves as a dynamic information revelation mechanism. Our model shares the feature that information structures are chosen dynamically, but in our setting the designer chooses the *architecture* of information processing rather than the informativeness of signals per se.

The common thread across these papers is that information is a flow variable that must be managed over time. Our contribution is to study how this flow is governed by organizational structure: in a hierarchy with attention-constrained agents, the optimal architecture determines not only *what* information is transmitted but also *how much* of it survives at each stage.

2.4 AI and Organizations

A substantial empirical and theoretical literature documents the effects of artificial intelligence on productivity, labor markets, and the organization of work. Our paper complements this literature by examining the *internal* structure of AI systems—how multiple AI agents should be organized into hierarchical architectures—rather than the effects of AI on human labor markets.

Acemoglu and Restrepo (2018) develop a task-based framework in which automation technologies displace labor in tasks previously performed by workers. Their key insight is that while automation directly reduces labor demand, it also indirectly encourages the creation of new tasks in which labor has a comparative advantage. Acemoglu and Restrepo (2022) extend this framework to study wage inequality, showing that a substantial

fraction of the rise in US wage inequality since 1980 can be attributed to automation of routine tasks. Our paper takes a different perspective: rather than studying how AI substitutes for or complements human labor in existing organizational structures, we ask how AI agents themselves should be organized into efficient architectures for processing information.

Acemoglu, Autor, Hazell, and Restrepo (2022) study the impact of AI on labor markets using evidence from online vacancies. They document that AI-related skills are increasingly demanded in occupations that combine cognitive and technical tasks, and that AI adoption is associated with shifts in skill requirements rather than pure displacement.

Noy and Zhang (2023) provide experimental evidence that generative AI tools substantially increase worker productivity. In a randomized controlled trial with professional writers, they find that access to ChatGPT reduces task completion time by 40% and increases output quality. Importantly, the productivity gains are largest for less-skilled workers, suggesting that AI can compress skill-based wage differentials. Peng et al. (2023) report similar findings for software developers using GitHub Copilot, documenting a 55.8% reduction in task completion time.

Ide and Talamas (2025) embed AI into the knowledge-hierarchy framework of Garicano (2000) but without attention scarcity. Klein and Wiecek (2026) study AI-driven organizational flattening but abstract from incentive constraints. We directly address the gap between AI capabilities (expanding context windows) and organizational limits (compounding attention loss).

Our work is complementary to this literature in the following sense. The existing literature studies the *macro* effects of AI adoption on human workers and firms—how AI changes the demand for skills, the distribution of wages, and the productivity of existing organizational forms. In contrast, we study the *micro* design of AI systems themselves: given that AI agents have heterogeneous capabilities and attention constraints, what is the optimal hierarchical architecture for information processing?

2.5 Information Design

The information design literature provides tools for modeling strategic communication. Kamenica and Gentzkow (2011) introduce Bayesian persuasion, where a sender designs a signal to influence a receiver’s beliefs. Lien, Qiu, and Zhong (2019) extend this to multi-layer communication networks. Our model differs in that the “design” is not strategic but constrained by attention scarcity.

Table 1 summarizes how our work relates to the most closely connected papers in each literature.

3 Model Setup

3.1 Overview: From Capital Depreciation to Information Depreciation

The central premise of this paper is that information, like physical capital, depreciates as it flows through an organizational chain. Each processing layer—each algorithmic transformation, each forwarding step, each summarization—introduces noise, compression, and loss. The rate of this *information depreciation* is not purely exogenous; it is endogenously

Table 1: Literature Mapping

Contribution	Relation to Our Model	Differentiation
Garicano (2000) Dessein and Santos (2006)	Knowledge hierarchy Organizational attention	+ attention scarcity + continuous allocation + chain degradation
Hadfield-Menell (2021) Phelps and Ranson (2023) Yang and Zhai (2025)	Single-layer PA in AI Behavioral validation with LLMs Heuristic: coordination costs	+ chain extension + cascade mechanism + multi-layer experimental design Formal proof: alignment costs persist
Tomašev, Paquet, and Tuyls (2026)	Bounded rationality in chains	Incentive misalignment in chains
Klein and Wieczorek (2026)	Coordination flattening	Attention scarcity bounds depth

determined by how attention is allocated across the upstream signals at each layer. Our model treats the delegation chain as a sequence of information-processing stages in which the scarce resource is attention capacity, and the central organizational problem is to design an attention architecture that maximizes the value of the final output.

This framing builds on a direct analogy to capital theory. In the neoclassical growth model, capital depreciates at rate δ per period: $K_{t+1} = (1 - \delta)K_t + I_t$. In our model, signal precision depreciates at rate $(1 - \eta)$ per layer:

$$\tau_{\ell+1,k}^0 = \eta \cdot \tau_{\ell,k}^0, \quad \eta \in (0, 1), \quad (1)$$

where η is the *information depreciation factor*. Just as the depreciation rate δ determines how much capital must be replaced to maintain productivity, η determines how much attention must be invested at each layer to preserve signal quality. The optimal depth of the chain is determined by the point at which the marginal attention cost of adding another layer exceeds the marginal information gain from deeper processing.

This perspective differs fundamentally from the classical principal-agent framework à la Holmström and Milgrom (1991). In the HM model, the agent chooses a scalar effort level e that maps into output through a production function $q = f(e) + \varepsilon$. The distortion is moral hazard: the agent would prefer to set $e = 0$. In our framework, the agent chooses a vector of attention weights $\alpha = (\alpha_1, \dots, \alpha_K)$ subject to a budget constraint $\sum_k \alpha_k \leq A$. The distortion is information depreciation compounded by rational inattention: even if the agent fully exhausts the budget A , the finite capacity of each layer implies that signals are processed with endogenously determined loss. The agent is not shirking; she faces a technological constraint on how much information can be preserved.

This distinction has three immediate implications for the study of AI delegation chains.

First, it provides a direct micro-foundation for the attenuation of signal quality across

layers. The information depreciation factor η captures, in reduced form, the algorithmic processes—tokenization, embedding projection, attention-weight normalization, and feed-forward transformations—that degrade signal fidelity in each forward pass of a transformer. Rather than treating precision as exogenous or arbitrarily layer-dependent, we derive its evolution from a primitive structural parameter.

Second, it endogenizes the effective precision parameter. In the HM framework, the signal precision τ is an exogenous parameter that the principal might manipulate through incentives. In our framework, the effective precision τ_ℓ^* at layer ℓ emerges from the interplay of three primitives: the information depreciation factor η , the attention budget A_ℓ , and the attention technology $g_\ell(\cdot)$. The principal influences τ_ℓ^* not by writing contracts but by designing the attention architecture—choosing the depth of the chain, the width of context windows, and the allocation of processing capacity across layers.

Third, it structures the output function as the outcome of an information-transmission technology. Equation (5) below derives precision from attention allocation given depreciated signal quality. The production function is not assumed; it follows from the technology of selective attention applied to successively degraded signals.

Relationship to Transformer Architectures. Our notion of an “attention budget” A_ℓ is deliberately stylized. It is most directly interpretable as a simplified representation of the *context-window length* of a transformer-based LLM—the maximum number of tokens that can be processed in a single forward pass (e.g., 4,096 tokens for GPT-3.5, 128,000 for GPT-4, or 1,000,000 for Gemini 1.5 Pro). However, we emphasize that this mapping is a deliberate simplification. Actual transformer attention involves *soft* allocation through learned query-key-value interactions, *multi-head* parallel processing, and *causal masking* constraints that make the effective attention pattern highly non-linear and context-dependent. Our linear budget constraint (4) abstracts from these complexities to deliver analytical tractability. We return to this limitation in the discussion below.

Our framework builds on the rational-inattention literature initiated by Sims (2003, 2006) and its organizational extension by Dessein and Santos (2006) and Dessein, Galeotti, and Santos (2016). The key departure from Sims is that our agents operate in a hierarchy where the output of one agent’s attention-allocation problem becomes the (depreciated) input to the next agent’s problem. The key departure from Dessein–Santos is that our agents are artificial—their attention budgets, costs, and signal structures are directly measurable from API logs—which allows us to calibrate and test the model with unprecedented precision.

Limitations and Scope Conditions. We acknowledge two important limitations of our stylized representation. First, the linear budget constraint (4) does not capture the non-linearities of actual transformer attention, where the effective processing capacity depends on the interaction between query complexity, key dimensionality, and the attention-head structure. A more micro-founded approach would model attention through the softmax operation directly; we sacrifice this realism for analytical clarity. Second, the information depreciation factor η is assumed constant across layers and signals. In practice, different types of signals (text, numerical, structured) may depreciate at different rates, and the depreciation rate may itself depend on the attention allocation (more attention may better preserve signal quality). These extensions are discussed in Section 8.

3.2 The Environment

Assumption 1 (Hierarchy). *There are $L + 1$ layers indexed by $\ell = 0, 1, \dots, L$. Layer 0 is the principal (human or top-level AI) who sets the organizational architecture. Layers $\ell = 1, \dots, L$ are AI agents arranged in a chain: the output of layer $\ell - 1$ serves as the input to layer ℓ . Layer L is the final executor that produces the observable output.*

Assumption 2 (State and Signals with Information Depreciation). *The underlying state is $\omega \in \Omega \subseteq \mathbb{R}^d$, representing the correct response to a complex task (e.g., the correct answer to a coding problem, the accurate diagnosis from medical records, or the valid proof of a mathematical theorem). At each layer ℓ , the agent receives K_ℓ signals $\{s_{\ell,1}, \dots, s_{\ell,K_\ell}\}$ from the layer above. Each signal is a noisy projection of the state:*

$$s_{\ell,k} = H_{\ell,k}\omega + \varepsilon_{\ell,k}, \quad \varepsilon_{\ell,k} \sim \mathcal{N}(0, \sigma_{\ell,k}^2 I), \quad (2)$$

where $H_{\ell,k} \in \mathbb{R}^{d_k \times d}$ is a linear observation operator.

The intrinsic precision of signal k at layer ℓ is:

$$\tau_{\ell,k}^0 = \bar{\tau}_k^0 \cdot \eta^\ell, \quad \eta \in (0, 1), \quad (3)$$

where $\bar{\tau}_k^0 \equiv 1/\bar{\sigma}_k^2$ is the raw precision of signal k at the source (layer 0), and η is the information depreciation factor.

The signal structure in Assumption 2 captures the empirical reality of AI delegation chains. In a retrieval-augmented generation (RAG) pipeline, the signals are retrieved documents; in a multi-agent coding system, they are code snippets and error messages from upstream agents; in a hierarchical reasoning chain, they are intermediate conclusions. The observation operator $H_{\ell,k}$ captures the fact that each signal may reveal only a subset of the state.

The information depreciation equation (3) is a core structural assumption. It posits that every layer through which a signal passes reduces its intrinsic precision by a constant factor $\eta < 1$. The economic rationale for $\eta < 1$ is threefold:

1. **Algorithmic noise.** Each forward pass of a transformer introduces stochasticity through dropout, layer normalization, and floating-point operations.
2. **Compression loss.** High-dimensional signal representations are projected into lower-dimensional spaces, discarding fine-grained information.
3. **Cumulated processing error.** Each layer’s output is an imperfect summary of its inputs; these imperfections compound through the chain.

The assumption that η is constant across layers and signals is a simplification; we discuss heterogeneous depreciation in Section 8.

Remark 1 (Scalar-State Simplification). *In the baseline analysis, we set $d = 1$ (scalar state) and $H_{\ell,k} = 1$ for all ℓ, k . This simplifies the precision-aggregation technology to scalar addition: $\tau_\ell = \tau_\ell^{\text{prior}} + \sum_k g_\ell(\alpha_{\ell,k})\tau_{\ell,k}^0$. The scalar assumption is restrictive when signals reveal different dimensions of a multi-faceted state. In Section 8, we sketch how the model extends to $d > 1$ with a precision matrix; the key insight—that attention scarcity bounds depth through compounding precision loss—is preserved.*

Assumption 3 (Attention Budget). *Each agent at layer ℓ has a finite attention budget $A_\ell > 0$. The agent chooses attention weights $\boldsymbol{\alpha}_\ell = (\alpha_{\ell,1}, \dots, \alpha_{\ell,K_\ell})$ satisfying*

$$\sum_{k=1}^{K_\ell} \alpha_{\ell,k} \leq A_\ell, \quad \alpha_{\ell,k} \geq 0 \quad \forall k. \quad (4)$$

The weight $\alpha_{\ell,k}$ measures the fraction of the agent’s attention capacity devoted to signal k . If $\alpha_{\ell,k} = 0$, the signal is completely ignored.

The attention budget A_ℓ is the central scarce resource in our model. In transformer-based LLMs, it has three direct interpretations: (i) the context-window length; (ii) the number of attention heads; and (iii) the effective working-memory capacity. The budget constraint (4) is deliberately stylized: it captures the fundamental scarcity of processing capacity in a tractable linear form, even though actual transformer attention involves non-linear softmax interactions and multi-head parallelism. We view the linear budget as a first-order approximation that preserves the key economic insight—scarcity forces selectivity—while admitting closed-form solutions.

It is important to emphasize that A_ℓ is not a choice variable of the agent at layer ℓ ; it is a technological parameter determined by the model architecture. The principal chooses A_ℓ when selecting which model to deploy at each layer. This creates a two-stage organizational design problem: the principal chooses the architecture $\{A_\ell\}_{\ell=1}^L$ and the chain depth L ; the agents then solve their attention-allocation problems given these budgets.

Assumption 4 (Attention Technology). *The effective precision achieved by the agent at layer ℓ is*

$$\tau_\ell(\boldsymbol{\alpha}_\ell) = \sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k}) \tau_{\ell,k}^0, \quad (5)$$

where $g_\ell : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a twice-differentiable, increasing, and concave function satisfying $g_\ell(0) = 0$, $g'_\ell(0) > 0$, and $\lim_{\alpha \rightarrow \infty} g'_\ell(\alpha) = 0$.

The function g_ℓ captures the attention technology: how does allocating attention weight α to a signal translate into effective processing precision? The concavity of g_ℓ reflects diminishing returns to attention. Our baseline specification uses a power function:

$$g_\ell(\alpha) = \alpha^\gamma, \quad \gamma \in (0, 1). \quad (6)$$

The parameter γ is the attention elasticity of precision. When γ is close to 1, attention is nearly linear in precision. When γ is small, attention is highly concave. We calibrate γ from empirical data on LLM accuracy as a function of context length in Section 6.

Remark 2 (Robustness of the $g(\cdot)$ Specification). *The qualitative properties of our model—the existence of an interior optimal depth, the comparative statics with respect to η and A_ℓ , and the efficiency rationales for chaining—do not depend on the power-function specification in equation (6). Any g_ℓ satisfying the regularity conditions in Assumption 4 yields the same first-order insights because the key economic force is the concavity of g_ℓ combined with the linearity of the budget constraint. The power function is chosen for analytical tractability; alternative specifications (e.g., logarithmic, $g(\alpha) = \log(1 + \alpha)$; or bounded, $g(\alpha) = \alpha/(1 + \alpha)$) yield similar qualitative results but less transparent closed forms. We explore the sensitivity of our results to the functional form of g in Appendix ??.*

Assumption 5 (Cost of Attention). *The cost of maintaining attention capacity A_ℓ at layer ℓ is $c_\ell(A_\ell)$, where $c_\ell : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is twice-differentiable, increasing, and convex, with $c_\ell(0) = 0$ and $c'_\ell(0) > 0$.*

For our baseline analysis, we use a quadratic cost:

$$c_\ell(A) = \kappa_\ell A + \frac{\phi_\ell}{2} A^2, \quad (7)$$

where $\kappa_\ell > 0$ is the marginal cost of the first unit of attention and $\phi_\ell \geq 0$ captures the convexity of compute costs at scale.

3.3 The Agent’s Problem: Attention Allocation

Given the environment above, the agent at layer ℓ solves the following optimization problem:

$$\max_{\alpha_\ell : (4)} V_\ell(\tau_\ell(\alpha_\ell)) - c_\ell(A_\ell), \quad (8)$$

where $V_\ell(\tau)$ is the value of posterior precision τ for decision-making at layer ℓ .

The value function $V_\ell(\tau)$ depends on the decision problem facing the agent. In our baseline, we assume quadratic loss (normal–normal conjugate model), which yields:

$$V_\ell(\tau) = -\psi_\ell \text{Var}(\omega \mid \tau) = -\frac{\psi_\ell}{\tau}, \quad (9)$$

where $\psi_\ell > 0$ is the marginal value of accuracy. With normal signals and a normal prior, the posterior variance is $\text{Var}(\omega \mid \tau) = 1/\tau$, so $V_\ell(\tau) = -\psi_\ell/\tau$. Since τ enters negatively, maximizing V_ℓ is equivalent to maximizing τ .

More generally, $V_\ell(\tau)$ can incorporate non-quadratic payoffs, discrete actions, or strategic interactions with downstream agents. We explore these extensions in Section 8. For the single-layer analysis in Section 4, the precise functional form of V_ℓ matters only through its monotonicity and curvature.

3.4 Key Distinctions: Information Depreciation versus Moral Hazard

Before proceeding to the analysis, we pause to articulate three core distinctions between our information-depreciation framework and the Garicano–Holmström–Milgrom (GHM) framework.

Distinction 1: Information depreciation, not moral hazard, is the source of distortion. In the HM model, agents choose costly effort e to improve output quality. The distortion is moral hazard: agents prefer low effort, and incentives must be designed to induce the efficient level. In our model, distortion arises from the compounding depreciation of signal quality as information flows through the chain—at rate $(1 - \eta)$ per layer—and from the binding attention constraint that forces agents to selectively process only a subset of signals. The agent is not shirking; she faces a technological constraint on how much information can be preserved. The organizational design problem is to set the chain depth, the attention budgets, and the signal routing so that the marginal benefit of deeper processing equals the marginal cost of depreciation plus attention.

Distinction 2: Selective processing of depreciated signals, not shirking, generates inefficiency. In the HM model, inefficiency arises because the agent’s effort is unobservable and the agent under-supplies it relative to the first-best. In our model, inefficiency arises because (i) each layer depreciates signal quality by factor η and (ii) the agent, facing a finite budget A_ℓ , rationally allocates attention only to the signals with highest effective precision. The ignored signals are not random; they are the ones with low post-depreciation precision, low importance for the decision, or high opportunity cost of attention. This generates testable predictions: in an LLM pipeline, we should observe that models drop low-quality retrieved documents before high-quality ones, that the dropout rate increases with chain depth (because η^ℓ shrinks the effective precision of upstream signals), and that the quality of the final output depends more on the depth–attention trade-off than on any notion of “effort.”

Distinction 3: Organization design as information architecture, not incentive design. In the HM model, the principal designs incentives (wages, bonuses, monitoring) to align the agent’s effort with organizational goals. In our model, the principal designs the *information architecture*: which signals flow to which layers, how much context each layer receives (the attention budgets A_ℓ), how many layers the chain contains (the depth L), and how the outputs of one layer feed into the next. The design variables are $\{A_\ell\}$ (attention budgets), $\{K_\ell\}$ (signal counts), L (chain depth), and the routing structure—not wages or performance contracts. Our model complements the GHM framework by adding information-processing microfoundations: where GHM asks “how should tasks and incentives be structured to motivate agents?” we ask “how should attention and depth be structured to preserve information quality?” The two questions are orthogonal and jointly relevant for the design of AI delegation systems.

3.5 Summary of Notation

For reference, Table 2 summarizes the key symbols used in the model.

4 Single-Layer Analysis

In this section, we solve the attention-allocation problem for a single layer, derive the optimal attention weights, characterize the resulting distortion, and establish comparative statics with respect to the attention budget, signal importance, and signal heterogeneity. We then connect our results to the rational-inattention literature and derive two propositions that form the building blocks for the multi-layer analysis in Section 5.

The single-layer problem is not merely a stepping stone. It is the core economic mechanism of the paper: everything that happens in a delegation chain—distortion, error propagation, depth optimization—traces back to how each individual layer allocates its scarce attention.

4.1 The Optimization Problem

Consider layer ℓ in isolation. The agent faces K_ℓ signals with intrinsic precisions $\{\tau_{\ell,k}^0\}_{k=1}^{K_\ell}$ and chooses attention weights $\boldsymbol{\alpha}_\ell = (\alpha_{\ell,1}, \dots, \alpha_{\ell,K_\ell})$ subject to the budget constraint (4). The effective precision is given by (5), and the agent’s payoff is $V_\ell(\tau_\ell(\boldsymbol{\alpha}_\ell)) - c_\ell(A_\ell)$. Since

Table 2: Notation Summary

Symbol	Definition
ℓ	Layer index, $\ell = 0, 1, \dots, L$
ω	Underlying state (correct answer / true state)
η	Information depreciation factor, $\eta \in (0, 1)$
$s_{\ell,k}$	Signal k at layer ℓ
K_ℓ	Number of signals at layer ℓ
$\bar{\tau}_k^0$	Raw intrinsic precision of signal k at source (layer 0)
$\tau_{\ell,k}^0$	Intrinsic precision of signal (ℓ, k) after ℓ layers of depreciation: $\tau_{\ell,k}^0 = \bar{\tau}_k^0 \cdot \eta^\ell$
A_ℓ	Attention budget at layer ℓ
$\alpha_{\ell,k}$	Attention weight on signal (ℓ, k)
$g_\ell(\cdot)$	Attention technology (precision transformation function)
$\tau_\ell(\boldsymbol{\alpha}_\ell)$	Effective precision at layer ℓ
ρ_ℓ	Endogenous precision ratio (relative to the precision that would obtain under full attention, as determined by the assumed functional form g_ℓ)
$V_\ell(\tau)$	Value of precision τ for decision-making
$c_\ell(A_\ell)$	Cost of attention capacity A_ℓ
γ	Attention elasticity of precision (power-function parameter)
κ_ℓ, ϕ_ℓ	Cost parameters (linear and quadratic)

$c_\ell(A_\ell)$ does not depend on α_ℓ , the optimization problem is:

$$\max_{\alpha_\ell : (4)} V_\ell \left(\sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k}) \tau_{\ell,k}^0 \right). \quad (10)$$

We assume throughout that $V_\ell(\tau)$ is increasing and concave in τ . This is satisfied by the quadratic-loss specification (9) and by most standard decision problems.

4.2 First-Order Conditions and Optimal Attention

Let $\lambda_\ell \geq 0$ be the Lagrange multiplier on the budget constraint (4). The Lagrangian is:

$$\mathcal{L}(\alpha_\ell, \lambda_\ell) = V_\ell \left(\sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k}) \tau_{\ell,k}^0 \right) - \lambda_\ell \left(\sum_{k=1}^{K_\ell} \alpha_{\ell,k} - A_\ell \right) + \sum_{k=1}^{K_\ell} \mu_{\ell,k} \alpha_{\ell,k}, \quad (11)$$

where $\mu_{\ell,k} \geq 0$ are multipliers on the non-negativity constraints.

The first-order condition for $\alpha_{\ell,k}$ is:

$$V'_\ell(\tau_\ell(\alpha_\ell)) \cdot g'_\ell(\alpha_{\ell,k}) \tau_{\ell,k}^0 = \lambda_\ell - \mu_{\ell,k}. \quad (12)$$

This condition has a straightforward economic interpretation. The left-hand side is the marginal benefit of allocating attention to signal k : the marginal value of precision (V'_ℓ) times the marginal precision gain from attention ($g'_\ell(\alpha_{\ell,k}) \tau_{\ell,k}^0$). The right-hand side is the marginal cost: the shadow price of the attention budget (λ_ℓ) minus the relaxation of the non-negativity constraint ($\mu_{\ell,k}$). At an interior solution ($\alpha_{\ell,k} > 0$), $\mu_{\ell,k} = 0$ and the marginal benefit equals the shadow price λ_ℓ .

Proposition 1 (Optimal Attention Allocation). *Suppose V_ℓ is increasing and concave, g_ℓ is increasing and concave with $g'_\ell(0) > 0$, and $\tau_{\ell,k}^0 > 0$ for all k . Then:*

(i) *The optimal attention allocation $\alpha_\ell^*(A_\ell)$ exists and is unique.*

(ii) *If the budget binds ($\sum_k \alpha_{\ell,k}^* = A_\ell$), the optimal weights satisfy*

$$g'_\ell(\alpha_{\ell,k}^*) \tau_{\ell,k}^0 = g'_\ell(\alpha_{\ell,j}^*) \tau_{\ell,j}^0 \quad \forall k, j \text{ with } \alpha_{\ell,k}^*, \alpha_{\ell,j}^* > 0. \quad (13)$$

(iii) *The optimal weight on signal k is increasing in its intrinsic precision $\tau_{\ell,k}^0$ and decreasing in the precision of other signals.*

Proof. See Appendix A. □

Intuition. The optimality condition (13) says that the agent equalizes the marginal precision product across all signals that receive positive attention. Signals with higher intrinsic precision receive more attention not because they are “preferred” in any absolute sense, but because they yield higher marginal returns: each unit of attention devoted to a high-quality signal extracts more information than a unit devoted to a low-quality signal. This is the familiar logic of equalizing marginal products across inputs, applied here to the allocation of attention rather than labor or capital.

For the power-function specification $g_\ell(\alpha) = \alpha^\gamma$, the first-order condition becomes:

$$\alpha_{\ell,k}^* = \left(\frac{\gamma \tau_{\ell,k}^0}{\lambda_\ell} \right)^{\frac{1}{1-\gamma}} \quad (\text{interior solutions}). \quad (14)$$

This closed form reveals the key comparative static: the optimal attention weight is log-convex in intrinsic precision when $\gamma < 1/2$ and log-concave when $\gamma > 1/2$. In the empirically relevant case of $\gamma \approx 0.3$ (calibrated from LLM accuracy data), small differences in signal quality are amplified into large differences in attention allocation: a signal with twice the precision receives approximately $2^{1/(1-0.3)} \approx 2.69$ times the attention. This amplification is the mechanism by which “important” signals dominate and “unimportant” signals are starved.

4.3 Distortion and the Attention Gap

We now define the distortion arising from attention allocation and characterize how it depends on the attention budget and signal structure.

Definition 1 (Attentional Distortion). *The distortion on signal k at layer ℓ is*

$$\delta_{\ell,k} = (\bar{\alpha}_{\ell,k} - \alpha_{\ell,k}^*)^+, \quad (15)$$

where $\bar{\alpha}_{\ell,k}$ is the attention weight required to fully process signal k (i.e., achieve $\tau_{\ell,k}^0$ without loss), and $(\cdot)^+ = \max\{\cdot, 0\}$.

In the baseline where $\bar{\alpha}_{\ell,k} = 1$ for all k (unit attention achieves full precision), the distortion is simply $\delta_{\ell,k} = (1 - \alpha_{\ell,k}^*)^+$. If $\alpha_{\ell,k}^* \geq 1$, the signal is fully processed and $\delta_{\ell,k} = 0$. If $\alpha_{\ell,k}^* < 1$, the signal is under-processed and the distortion is positive.

We can also define an aggregate distortion measure:

Definition 2 (Aggregate Distortion). *The aggregate attentional distortion at layer ℓ is*

$$\Delta_\ell = \frac{\sum_{k=1}^{K_\ell} \omega_{\ell,k} \delta_{\ell,k}}{\sum_{k=1}^{K_\ell} \omega_{\ell,k}}, \quad (16)$$

where $\omega_{\ell,k} > 0$ is the importance weight of signal k for the layer’s decision. In the quadratic-loss case, $\omega_{\ell,k} = \tau_{\ell,k}^0$.

Proposition 2 (Distortion Comparative Statics). *The aggregate distortion Δ_ℓ satisfies:*

- (i) Δ_ℓ is decreasing in the attention budget A_ℓ : $\partial\Delta_\ell/\partial A_\ell < 0$.
- (ii) Δ_ℓ is increasing in the number of signals K_ℓ : $\partial\Delta_\ell/\partial K_\ell > 0$ when A_ℓ is held fixed.
- (iii) Δ_ℓ is decreasing in the heterogeneity of signal precisions: holding mean precision fixed, Δ_ℓ is lower when $\{\tau_{\ell,k}^0\}$ is more dispersed.

Proof. See Appendix B. □

Intuition. Proposition 2 delivers three testable predictions. First, increasing the attention budget reduces distortion: if the principal upgrades from GPT-3.5 (4K context) to GPT-4 (128K context), the model can attend to more signals and distortion falls. Second, increasing the number of signals while holding the budget fixed increases distortion: this is the mechanism behind “attention overload” in RAG systems. Third, signal heterogeneity reduces aggregate distortion because the optimal allocation concentrates attention on the best signals. If all signals are of equal quality, the agent spreads attention evenly and all signals are under-processed. If signals vary in quality, the agent concentrates on the high-quality ones, achieving near-full precision on the most important signals while deliberately sacrificing the low-quality ones.

4.4 The Endogenous Distortion Factor

We now derive the endogenous distortion measure that connects the single-layer attention-allocation problem to the multi-layer transmission analysis.

Definition 3 (Single-Layer Distortion Measure). *The single-layer distortion measure at layer ℓ is*

$$\tilde{\delta}_\ell = 1 - \Delta_\ell = 1 - \frac{\sum_k \omega_{\ell,k} (\bar{\alpha}_{\ell,k} - \alpha_{\ell,k}^*)^+}{\sum_k \omega_{\ell,k}}. \quad (17)$$

The measure $\tilde{\delta}_\ell \in [0, 1]$ captures the fraction of signal importance that survives the attention-allocation process at a single layer, ignoring upstream effects. When $\tilde{\delta}_\ell = 1$, all signals are fully processed and no distortion occurs. When $\tilde{\delta}_\ell = 0$, all signals are ignored and the layer transmits pure noise. In general, $\tilde{\delta}_\ell$ is a function of the attention budget, the number of signals, their precisions, and the attention technology:

$$\tilde{\delta}_\ell = \tilde{\delta}(A_\ell, K_\ell, \{\tau_{\ell,k}^0\}, g_\ell). \quad (18)$$

This is the core endogenization result. The distortion measure is not a primitive parameter but a derived object that depends on observable technological and organizational variables.

Remark 3 (From $\tilde{\delta}_\ell$ to ρ_ℓ). *The single-layer measure $\tilde{\delta}_\ell$ is useful for analyzing distortion at an isolated layer. In the multi-layer analysis (Section 5), we introduce the endogenous transmission factor $\rho_\ell \equiv \tau_\ell^* / \tau_\ell^{FB}$, which accounts for the compounding of inherited precision. For the bottom layer with $\tau_0^{\text{prior}} = 0$, the two coincide: $\rho_0 = \tilde{\delta}_0 = 1 - \Delta_0$. For upper layers, ρ_ℓ generalizes $\tilde{\delta}_\ell$ by incorporating the prior precision.*

4.5 Connection to Rational Inattention

Our model is a direct extension of the rational-inattention framework to hierarchical organizations. We now make this connection explicit.

Sims (2003): The single-agent benchmark. In Sims’s model, a single agent faces a multidimensional state $\omega \in \mathbb{R}^d$ and chooses an information structure subject to a mutual-information constraint $I(\omega; \hat{\omega}) \leq \kappa$. Our model specializes Sims in two directions and generalizes it in one. We specialize by assuming a linear precision-aggregation technology (Equation (5)) rather than the general information-structure choice. We specialize by assuming a hard attention budget rather than an entropy-based information-cost function. We generalize by embedding the single-agent problem in a hierarchy where the output of one agent becomes the input to the next.

The hard budget vs. entropy-cost distinction is important. Sims’s entropy cost is analytically elegant, but it is less directly mapped to AI technology. The mutual information $I(\omega; \hat{\omega})$ is difficult to measure in an LLM pipeline, whereas the context-window length A_ℓ is directly observable. Our hard-budget formulation trades some analytical elegance for empirical tractability.

Dessein and Santos (2006): Organizational attention. Dessein and Santos model an organization in which agents have a fixed attention capacity that can be used for both sending and receiving information. Our model extends theirs in three ways. First, we allow attention to be continuously allocated across signals rather than discretely assigned to tasks. Second, we allow the attention budget to vary across layers (A_ℓ need

not equal $A_{\ell'}$), capturing the fact that different LLMs in a chain have different context-window sizes. Third, we model the endogenous degradation of information through the chain.

Dessein, Galeotti, and Santos (2016): Division of labor. In their follow-up paper, Dessein et al. study how attention constraints generate specialization. Our Proposition 2(iii) echoes this result: heterogeneity in signal quality leads to specialization in attention allocation. We extend their analysis by deriving the optimal chain depth from attention costs.

Limitations of the microfoundation. Our model makes two deliberate simplifications relative to actual transformer architectures. First, the linear budget constraint (4) is a stylized representation of the context window limit; real transformers use soft attention weights rather than hard capacity constraints. Second, we abstract from the token-token interaction structure of self-attention, treating each signal as an independent information source. These simplifications trade descriptive realism for analytical tractability and should be interpreted as a first-order approximation rather than a literal mapping.

4.6 Alternative: Entropy Cost Formulation

4.6.1 Entropy Cost Setup

In the rational inattention framework of Sims (2003), the agent’s information processing cost is proportional to the reduction in uncertainty (measured by mutual information) achieved by observing a signal. Specifically, if the agent chooses an information structure $P(\tilde{s} \mid \omega)$ that maps the true state $\omega \sim \mathcal{N}(\mu, \tau^{-1})$ into a signal realization $\tilde{s}$, the associated cost is:

$$C(I) = \theta \cdot I(\omega; \tilde{s}), \quad (19)$$

where $\theta > 0$ is the marginal price of information and $I(\omega; \tilde{s})$ denotes the mutual information:

$$I(\omega; \tilde{s}) = h(\omega) - h(\omega \mid \tilde{s}) = \frac{1}{2} \log_2 \left(\frac{\tau_{\text{post}}}{\tau} \right), \quad (20)$$

where $\tau_{\text{post}} = \tau + \tau_s$ is the posterior precision after observing signal $\tilde{s} = \omega + \varepsilon$ with noise precision τ_s , and $h(\cdot)$ denotes differential entropy.

When a chain of L agents each face entropy costs, agent ℓ chooses a signal structure $P(\tilde{s}_\ell \mid \omega)$ at cost $\theta_\ell \cdot I(\omega; \tilde{s}_\ell)$, and then passes a processed message m_ℓ to the next layer. The key difference from the hard-budget formulation is that *all signals are processed, but with endogenously varying precision*, rather than a subset being selected with perfect precision.

4.6.2 Entropy-Cost Version of Our Model

Consider our multi-layer information processing chain under entropy costs. Each layer ℓ chooses the precision τ_ℓ of its signal subject to an entropy cost $\theta_\ell \cdot I(\omega; \tilde{s}_\ell)$. The total cost for the chain is:

$$C_{\text{total}} = \sum_{\ell=1}^L \theta_\ell \cdot I(\omega; \tilde{s}_\ell) = \sum_{\ell=1}^L \frac{\theta_\ell}{2} \log \left(1 + \frac{\tau_\ell}{\tau_{\ell-1}^*} \right), \quad (21)$$

where $\tau_{\ell-1}^*$ is the effective precision inherited from the previous layer. The optimization problem at each layer is to choose τ_ℓ to maximize:

$$U_\ell(\tau_\ell) = V(\tau_\ell^*) - \theta_\ell \cdot I(\omega; \tilde{s}_\ell), \quad (22)$$

where $\tau_\ell^* = \tau_{\ell-1}^* + \tau_\ell$ is the posterior precision.

Key Result: Robustness of Finite Depth. We now establish that the finite-optimal-depth result carries over to the entropy-cost setting:

Proposition 3 (Finite Optimal Depth under Entropy Costs). *Suppose each layer ℓ faces an entropy cost with parameter $\theta_\ell > 0$, and the value function $V(\tau)$ satisfies $V'(\tau) > 0$, $V''(\tau) < 0$, and the Inada-like condition $\lim_{\tau \rightarrow \infty} V'(\tau) = 0$. Then:*

- (i) **Concave precision response:** *The optimal precision $\tau_\ell^*(\theta_\ell)$ chosen at each layer satisfies $\frac{\partial^2 \tau_\ell^*}{\partial \theta_\ell^2} > 0$, i.e., precision decreases convexly in the cost parameter.*
- (ii) **Diminishing information increment:** *The marginal contribution of layer ℓ to total precision, $\Delta_\ell \equiv \tau_\ell^* - \tau_{\ell-1}^*$, is decreasing in ℓ when θ_ℓ is non-decreasing.*
- (iii) **Finite optimal depth:** *If $\inf_\ell \theta_\ell > 0$, there exists a finite $L^* < \infty$ such that adding layer $L^* + 1$ yields negative net value.*

Proof Sketch. (i) At layer ℓ , the first-order condition for optimal precision is:

$$V'(\tau_{\ell-1}^* + \tau_\ell) = \frac{\theta_\ell}{2} \cdot \frac{1}{\tau_{\ell-1}^* + \tau_\ell}. \quad (23)$$

This implicitly defines τ_ℓ^* as a function of θ_ℓ . By the implicit function theorem and the convexity of the entropy cost in precision, the response function is concave in $1/\theta_\ell$.

(ii) Since τ_ℓ^* is increasing in inherited precision $\tau_{\ell-1}^*$ but with slope less than 1 (due to $V'' < 0$), the increment Δ_ℓ decreases along the chain when cost parameters are bounded away from zero.

(iii) The net value of adding layer $L + 1$ is $V(\tau_{L+1}^*) - V(\tau_L^*) - \theta_{L+1} \cdot I(\omega; \tilde{s}_{L+1})$. Since $V'(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$ while the marginal entropy cost remains bounded below by $\frac{\theta}{(2\tau)^{-1}} > 0$, there exists a finite threshold beyond which the cost exceeds the benefit. $\square$

Intuition for Robustness. The critical insight is that the entropy cost function’s *convexity* in precision (arising from the logarithmic form of mutual information) naturally generates the same diminishing-returns property that our hard-budget model captures through the fixed-capacity constraint.

4.6.3 Discussion: Why Hard Budgets in the Baseline Model

We adopt the hard-budget assumption as our baseline for three substantive reasons related to *empirical operability* and the specific application to large language models:

Direct observability. The hard budget $\bar{A}_\ell$ maps directly and transparently to a measurable technological parameter: the context window length of an LLM (e.g., 4,096 tokens for GPT-3, 128,000 tokens for GPT-4 Turbo). This observable anchor allows us to discipline the model with empirical moments and conduct counterfactuals about window-size expansion.

Table 3: Comparison: Hard-Budget vs. Entropy-Cost Assumptions

Dimension	Hard-Budget	Entropy-Cost
Information loss	Signals beyond budget $\bar{A}_\ell$ ignored or truncated.	All signals processed with endogenously varying precision.
Optimization variable	Attention weight vector $\alpha_\ell \in \Delta^K$ (discrete capacity).	Information structure $P(\tilde{s} \mid \omega)$ (continuous precision).
Observability	$\bar{A}_\ell$ directly observable (context window size).	θ_ℓ latent; requires calibration.
LLM mapping	Direct: context window imposes hard token bound.	Indirect: entropy reduction to FLOPs mapping.
Core conclusion	Finite optimal depth (Prop. 8).	Finite optimal depth (Prop. 3).
Parametrization	Sharp cutoff at $\bar{A}_\ell$.	Smooth; no cutoff.
Applicable scenarios	Fixed context-window LLMs.	Variable-compute systems.

Descriptive accuracy for current LLM architectures. Current production LLMs operate with fixed context windows: tokens beyond the window length are either truncated entirely or compressed through external summarization mechanisms. This is precisely the hard-budget mechanism.

Analytical tractability. The hard-budget formulation yields a finite-dimensional optimization over the simplex Δ^K , enabling closed-form solutions for the attention allocation and sharp comparative statics.

Entropy costs as robustness check. We view the entropy-cost formulation primarily as a *robustness check* on the qualitative conclusions. As Proposition 3 demonstrates, the core mechanism—diminishing returns to depth driven by concave precision accumulation—is preserved across both formulations.

4.7 Propositions for the Multi-Layer Analysis

We conclude the single-layer analysis with two propositions that will serve as building blocks for the multi-layer analysis.

Proposition 4 (Precision Response to Budget). *Let $\tau_\ell^*(A_\ell) \equiv \tau_\ell(\alpha_\ell^*(A_\ell))$ be the optimal effective precision. Then:*

- (i) $\tau_\ell^*(A_\ell)$ is increasing and concave in A_ℓ .
- (ii) The marginal return to attention, $\partial \tau_\ell^* / \partial A_\ell$, is decreasing in K_ℓ .
- (iii) $\tau_\ell^*(A_\ell)$ is bounded above by $\bar{\tau}_\ell \equiv \sum_k \tau_{\ell,k}^0$.

Proof. See Appendix C. □

Proposition 5 (Cost-Minimizing Budget). *For a target precision level $\bar{\tau}$, the cost-minimizing attention budget $A_\ell^*(\bar{\tau})$ satisfies:*

- (i) $A_\ell^*(\bar{\tau})$ is increasing and convex in $\bar{\tau}$.
- (ii) The marginal cost of precision, $\partial c_\ell(A_\ell^*) / \partial \bar{\tau}$, is increasing in K_ℓ .

(iii) If $\bar{\tau} > \tau_\ell^*(K_\ell)$, then $A_\ell^*(\bar{\tau})$ does not exist: the target is infeasible.

Proof. See Appendix D. □

4.8 Discussion: Why This Is Not “Concave Benefit + Convex Cost”

We close this section by addressing a potential objection: is the attention-allocation model merely a relabeled version of the “concave benefit + convex cost = finite optimum” structure? We argue that it is not, for three reasons.

First, the concavity emerges from the technology, not the objective. In the standard framework, the principal assumes a concave benefit function and a convex cost function to ensure a finite optimum. In our framework, the concavity of $\tau_\ell^*(A_\ell)$ (Proposition 4) is *derived* from the attention technology: diminishing returns arise because the agent allocates attention to signals in order of quality, not because we assumed a concave production function.

Second, the optimization is over a vector, not a scalar. The HM model optimizes over a scalar effort e . Our model optimizes over a vector α_ℓ subject to a budget constraint. The solution is a system of K_ℓ equations with non-negativity constraints and a binding budget. The structure of the solution—which signals receive attention, which are ignored, and how the allocation changes with parameters—is economically meaningful and empirically testable.

Third, the distortion is endogenous and derived relative to the assumed $g_\ell(\cdot)$ technology. In the standard framework, the distortion parameter is exogenous and the output function is ad hoc. In our framework, the distortion measure $\tilde{\delta}_\ell$ is derived from the optimal attention allocation (Definition 3), the output function is the precision-aggregation technology (Equation (5)), and both are directly connected to observable AI technology.

Remark 4 (Robustness to functional form). *The qualitative comparative-statics conclusions (Propositions 2–5) do not depend on the specific functional form of $g_\ell(\cdot)$. All that is required is that g_ℓ be increasing and concave with $g'_\ell(0) > 0$. The power-function specification $g_\ell(\alpha) = \alpha^\gamma$ is adopted for analytical convenience and because it yields closed-form solutions; the same economic forces operate under any attention technology that exhibits diminishing marginal returns.*

4.9 Benchmark Model: The GHM Framework

To isolate the distinctive predictions of our attention-allocation model, we now construct and solve a benchmark formulation that nests our setting within the classical framework of Garicano (2000) and Holmström and Milgrom (1991) (henceforth, GHM).

4.9.1 Model Setup

Consider a chain of L identical agents indexed by $\ell = 1, \dots, L$. Each agent ℓ chooses a scalar effort level $e_\ell \in \mathbb{R}_+$, which represents the precision or quality of the signal that agent ℓ transmits upstream.

Assumption 6 (GHM Cost Structure). *The cost function ψ is twice continuously differentiable with:*

- (i) $\psi(0) = 0$, $\psi'(e) > 0$ for all $e \geq 0$;
- (ii) $\psi''(e) > 0$ for all $e \geq 0$ (strict convexity);
- (iii) $\lim_{e \rightarrow \infty} \psi'(e) = \infty$ (Inada condition).

Assumption 7 (GHM Production Technology). *The final output quality is given by*

$$q_L = f\left(\sum_{\ell=1}^L e_\ell\right) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad (24)$$

where $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is twice continuously differentiable with $f'(x) > 0$, $f''(x) < 0$ for all $x \geq 0$, and satisfies the Inada conditions $f'(0) = \infty$, $\lim_{x \rightarrow \infty} f'(x) = 0$.

Agent ℓ 's utility is:

$$U_\ell = \mathbb{E}[w(q_L)] - \psi(e_\ell) = a + b \cdot f\left(\sum_{j=1}^L e_j\right) - \psi(e_\ell). \quad (25)$$

4.9.2 Equilibrium Effort and Optimal Depth

Taking the depth L and incentive power b as given, each agent ℓ chooses e_ℓ to maximize (25). By symmetry, $e_\ell = e^*$ for all ℓ . The first-order condition is:

$$b \cdot f'(L \cdot e^*) = \psi'(e^*). \quad (26)$$

Proposition 6 (GHM Optimal Depth). *Suppose Assumptions 6–7 hold and the principal optimally sets $b^* > 0$. Then:*

- (i) *There exists a finite optimal depth $L_{GHM}^* < \infty$.*
- (ii) *As the marginal cost of effort vanishes—formally, as $\psi'(e) \rightarrow 0$ —the optimal depth diverges:*

$$\lim_{\kappa \rightarrow 0} L_{GHM}^*(\kappa) = \infty. \quad (27)$$

- (iii) *The equilibrium effort per layer satisfies $e^* \rightarrow \infty$ as $\kappa \rightarrow 0$.*

Proof. (i) At $L = 0$, the marginal benefit is strictly positive because $f'(0) = \infty$. As $L \rightarrow \infty$, $f'(L \cdot e^*) \rightarrow 0$ while $\psi(e^*) > 0$. By continuity, there exists at least one crossing.

(ii) Consider the family $\psi_\kappa(e) = \kappa \cdot \psi(e)$. For any fixed finite L , as $\kappa \rightarrow 0$, the FOC implies $f'(L \cdot e^*) = (\kappa/b)\psi'(e^*) \rightarrow 0$, which requires $L \cdot e^* \rightarrow \infty$. The marginal benefit is $O(1)$ while the marginal cost is $O(\kappa)$. Thus the principal always wants to add more layers, and $L_{GHM}^* \rightarrow \infty$. $\square$

4.9.3 Complementary Predictions

In our attention-allocation model, even as the per-unit cost of attention $\kappa \rightarrow 0$, each layer faces a *hard budget constraint* $\sum_k \alpha_{\ell,k} \leq A_\ell$. The optimal depth remains finite because:

- (a) The budget A_ℓ places an upper bound on effective precision.
- (b) The transmission factor $\rho_\ell = \tau_\ell^*/\tau_\ell^{\text{FB}} < 1$ is bounded away from 1 by the geometry of the budget constraint, regardless of κ .
- (c) The geometric decay of precision along the chain is driven by the *relative* allocation across signals, not by the absolute cost of attention.

Table 4: Model Predictions: Our Framework vs. GHM Benchmark

Prediction	Our Model	GHM Benchmark	Empirical Test
Finite optimal depth ($\kappa \rightarrow 0$)	Yes. Budget A_ℓ bounds precision regardless of cost.	No. $L_{\text{GHM}}^* \rightarrow \infty$ as $\psi'(e) \rightarrow 0$.	Observe depth response to API price cuts.
Budget allocation	Front-loaded. Attention concentrates at early layers.	Uniform. Identical effort e^* at all layers.	Compare model sizes across layers.
Signal heterogeneity	Reduces distortion. Selective attention raises ρ_ℓ .	No effect. Heterogeneity not in FOC.	Homogeneous vs. heterogeneous inputs.
Depth-precision relation	Geometric decay. $\tau_\ell \propto \rho^\ell$.	Arithmetic decay. $L \cdot e^*(L)$.	Error accumulation in chains.
Elasticity identification	Identifiable. From context-accuracy curves.	Not identifiable. e_ℓ unobservable.	Attention-weight data from APIs.

4.9.4 Four Dimensions of Complementarity

The predictions in Table 4 differ because the two models emphasize different economic primitives. GHM abstracts from attention allocation across heterogeneous signals, while our model abstracts from the team-production and incentive-design problems that GHM was built to analyze. The two frameworks are *complements*: each generates predictions the other does not.

1. Vector vs. scalar optimization. Our model generates additional predictions by incorporating *vector optimization*: each layer allocates a budget A_ℓ across multiple signals with heterogeneous precisions. This generates **selective neglect**—the optimal policy concentrates attention on the most informative signals and ignores others. GHM, by abstracting from signal heterogeneity, focuses on *scalar optimization* where the agent chooses a single effort level e_ℓ .

2. Endogenous vs. exogenous distortion. Our model generates additional predictions by making the transmission factor $\rho_\ell = \tau_\ell^*/\tau_\ell^{\text{FB}}$ **endogenous**: it depends on the signal structure, the attention technology $g_\ell(\cdot)$, and the budget A_ℓ . GHM abstracts from this channel; any distortion in that framework is **exogenous**—it would appear as a fixed parameter in the production function f .

3. Observable vs. unobservable inputs. The attention weights $\alpha_{\ell,k}$ in our model capture, in reduced form, how the model processes different information sources. Modern transformer architectures expose attention maps as byproducts of computation (Vaswani et al. 2017), which enables direct structural estimation. GHM, by focusing on moral hazard, treats effort e_ℓ as classically **unobservable**.

4. Behavior as $\kappa \rightarrow 0$: a complementary prediction. Proposition 6 establishes that $L_{\text{GHM}}^* \rightarrow \infty$ when $\psi'(e) \rightarrow 0$. Our model generates the complementary prediction that $L^* < \infty$ *even in the limit* because the budget constraint A_ℓ creates a hard bound on effective precision. Recent API price reductions represent a shock to κ . Under GHM,

firms should respond by deepening chains indefinitely; under our model, depth should stabilize. The evidence in Section 7 supports the latter prediction.

Remark 5 (What GHM does better). *The GHM framework naturally accommodates team production with free-riding, optimal incentive design through the choice of b , and rich contracting between principal and agents. Our model abstracts from these classical moral-hazard considerations to focus on the novel margin of attention allocation. A fully satisfactory theory would embed our attention-allocation problem within a GHM-style contracting framework; we leave this synthesis to future work.*

5 Multi-Layer Analysis and Optimal Depth

We now extend the analysis to multiple layers, derive the optimal organizational depth, and characterize the optimal budget allocation.

5.1 The Endogenous Transmission Factor

Before analyzing information transmission across layers, we formally define the endogenous transmission factor and relate it to the single-layer distortion measure.

Recall from Definition 3 that the single-layer distortion measure is $\tilde{\delta}_\ell = 1 - \Delta_\ell$. In the multi-layer context, it is more convenient to work directly with the ratio of achieved posterior precision to first-best precision:

Definition 4 (Endogenous Transmission Factor). *The endogenous transmission factor from layer ℓ to layer $\ell + 1$ is*

$$\rho_\ell \equiv \frac{\tau_\ell^*}{\tau_\ell^{FB}} = \frac{\tau_\ell^{prior} + \sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k}^*) \tau_{\ell,k}^0}{\tau_\ell^{prior} + \sum_{k=1}^{K_\ell} \tau_{\ell,k}^0} \in [0, 1], \quad (28)$$

where τ_ℓ^{FB} is the first-best posterior precision and τ_ℓ^* is the achieved posterior precision under the optimal attention allocation.

The transmission factor ρ_ℓ measures the fraction of first-best precision that survives layer ℓ . When $\rho_\ell = 1$, all signals receive sufficient attention and no precision is lost. When $\rho_\ell < 1$, the attention budget forces the layer to ignore or under-process some signals, creating a precision leakage.

Remark 6 (From Exogenous to Endogenous). *In previous work, the transmission factor was an exogenous parameter $\tau \in [0, 1]$ that the modeler specified. In our framework, ρ_ℓ is endogenously determined by the attention-allocation problem: it depends on the budget A_ℓ , the number of signals K_ℓ , their precisions $\{\tau_{\ell,k}^0\}$, and the attention technology $g_\ell(\cdot)$. Importantly, ρ_ℓ is derived relative to the assumed attention technology $g_\ell(\cdot)$ —changing the functional form of g_ℓ would alter the numerical value of ρ_ℓ even holding all primitives fixed.*

Remark 7 (Technology Dependence and Qualitative Robustness). *The numerical value of ρ_ℓ depends on the specific attention technology $g_\ell(\cdot)$. Under the power-function specification $g_\ell(\alpha) = \alpha^\gamma$, a larger context window (higher A_ℓ) raises ρ_ℓ ; under an alternative technology, the quantitative response may differ. However, the qualitative conclusions of our analysis—that optimal depth is finite and that budget allocation is front-loaded—do not depend on the specific functional form of $g_\ell(\cdot)$. All that is required is that g_ℓ be increasing and concave with $g'_\ell(0) > 0$.*

5.2 Information Transmission Across Layers

In a chain with L layers, the posterior precision at layer ℓ depends on the attention allocation at layer $\ell - 1$ through the endogenous transmission factor $\rho_{\ell-1}$. The prior precision at layer ℓ is the achieved posterior precision at layer $\ell - 1$:

$$\tau_\ell^{\text{prior}} = \rho_{\ell-1} \cdot \tau_{\ell-1}^*(A_{\ell-1}), \quad \ell = 1, \dots, L. \quad (29)$$

With $\tau_0^{\text{prior}} \equiv \tau_0^0$ (the raw signal precision at the bottom layer), the posterior precision at layer ℓ follows the recursive Bayesian-update structure:

$$\tau_\ell^* = \tau_\ell^{\text{prior}} + \sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k}^*) \tau_{\ell,k}^0, \quad \text{with } \tau_\ell^{\text{prior}} = \rho_{\ell-1} \cdot \tau_{\ell-1}^*. \quad (30)$$

Solving the recursion yields the cumulative precision at the top layer:

$$\tau_L^* = \sum_{\ell=1}^L \sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k}^*) \tau_{\ell,k}^0 \cdot \prod_{j=\ell}^{L-1} \rho_j + \tau_0^0 \cdot \prod_{j=0}^{L-1} \rho_j. \quad (31)$$

Each layer ℓ contributes its local signal precision $g_\ell(\alpha_{\ell,k}^*) \tau_{\ell,k}^0$, weighted by the subsequent transmission losses $\prod_{j=\ell}^{L-1} \rho_j$.

Precision Dynamics under Information Depreciation. We now characterize the dynamics of the precision path $\{\tau_\ell^*\}$ under the information depreciation model introduced in Assumption 2. Recall that in this framework, the raw signal quality at layer ℓ is $\tau_{\ell,k}^0 = \tau_k^0 \cdot \eta^\ell$, where $\eta \in (0, 1)$ is the per-layer signal retention factor.

To build intuition, consider first what happens when $\eta = 1$ (no signal degradation). Under uniform technology ($g_\ell = g$, $\rho_\ell = \rho$ constant), Equation (30) becomes $\tau_\ell^* = \rho \tau_{\ell-1}^* + C$ for some constant $C > 0$, which converges to $\bar{\tau} = C/(1 - \rho) > 0$ as $\ell \rightarrow \infty$. In this case, the precision path converges to a *positive limit*: adding more layers continues to increase precision (albeit with diminishing returns), and infinite depth is optimal whenever API costs are below a threshold. The boundedness of depth, if any, comes solely from cost considerations, not from fundamental information limits.

When $\eta < 1$, the dynamics change qualitatively. The following proposition characterizes the precision path under the information depreciation model with uniform technology.

Proposition 7 (Precision Path Characterization). *Under the information depreciation model (Assumption 2 with $\eta \in (0, 1)$) and uniform technology ($g_\ell = g$, $c_\ell = c$, $K_\ell = K$, $\rho_\ell = \rho$ for all ℓ), define the composite signal contribution constant*

$$C \equiv \sum_{k=1}^K g(\alpha_k^*) \tau_k^0, \quad (32)$$

where α_k^* is the optimal attention allocation to source k . The equilibrium posterior precision at layer ℓ satisfies the recurrence

$$\tau_\ell^* = \rho \cdot \tau_{\ell-1}^* + C\eta^\ell, \quad \tau_0^* = \sum_{k=1}^K \tau_k^0. \quad (33)$$

The closed-form solution is

$$\tau_L^* = \rho^L \tau_0^* + \frac{C\eta(\eta^L - \rho^L)}{\eta - \rho} \quad \text{for } \eta \neq \rho, \quad (34)$$

and $\tau_L^* = \eta^L(\tau_0^* + CL)$ when $\eta = \rho$. Moreover:

(i) **If $\eta < \rho$:** τ_ℓ^* is hump-shaped. It increases for early layers, reaches a maximum at some finite $\ell^* \geq 1$, then decreases monotonically. The limiting precision is $\lim_{\ell \rightarrow \infty} \tau_\ell^* = 0$.

(ii) **If $\eta > \rho$:** τ_ℓ^* is eventually decreasing. It may increase initially depending on initial conditions, but eventually declines monotonically to 0.

(iii) **If $\eta = \rho$:** $\tau_\ell^* = \eta^\ell(\tau_0^* + C\ell)$, which is eventually decreasing to 0.

In all cases, $\lim_{\ell \rightarrow \infty} \tau_\ell^* = 0$.

Proof. Step 1: Derivation of the recurrence. Under uniform technology and Assumption 2, the new signal contribution at layer ℓ is $\sum_{k=1}^K g(\alpha_k^*) \tau_k^0 \eta^\ell = C\eta^\ell$, where C is defined in (32). Substituting into (30) yields (33).

Step 2: Closed-form solution. Equation (33) is a first-order linear non-homogeneous difference equation.

Case 1: $\eta \neq \rho$. The homogeneous solution is $\tau_\ell^{(h)} = A\rho^\ell$. Guessing a particular solution $\tau_\ell^{(p)} = B\eta^\ell$ and substituting:

$$B\eta^\ell = \rho B\eta^{\ell-1} + C\eta^\ell \implies B = \frac{C\eta}{\eta - \rho}.$$

Applying $\tau_0^* = A + B$ and solving for τ_L^* yields (34).

Case 2: $\eta = \rho$. The recurrence becomes $\tau_\ell^* = \eta\tau_{\ell-1}^* + C\eta^\ell$. Defining $u_\ell \equiv \tau_\ell^*/\eta^\ell$, we have $u_\ell = u_{\ell-1} + C$, which telescopes to $u_\ell = \tau_0^* + C\ell$. Hence $\tau_L^* = \eta^L(\tau_0^* + CL)$.

Step 3: Vanishing precision (all cases). Since $\rho, \eta \in (0, 1)$, both $\rho^L \rightarrow 0$ and $\eta^L \rightarrow 0$. In all three cases, $\lim_{L \rightarrow \infty} \tau_L^* = 0$.

Step 4: Shape characterization. From (33), the layer-to-layer change is

$$\Delta_\ell \equiv \tau_\ell^* - \tau_{\ell-1}^* = -(1 - \rho)\tau_{\ell-1}^* + C\eta^\ell. \quad (35)$$

Subcase (i): $\eta < \rho$. The ρ^ℓ term dominates the closed-form solution for large ℓ . Since $\rho > \eta$, the negative term $-(1 - \rho)\tau_{\ell-1}^*$ (which decays as $\rho^{\ell-1}$) eventually dominates the positive term $C\eta^\ell$ (which decays as η^ℓ). Thus $\Delta_\ell < 0$ for all $\ell \geq \bar{L}_1$. For small ℓ , however, the positive term may dominate if τ_0^* is small relative to $C\eta$, causing an initial increase. The path is therefore hump-shaped.

Subcase (ii): $\eta > \rho$. As $\ell \rightarrow \infty$, $\tau_{\ell-1}^* \sim C\eta^\ell/(\eta - \rho)$. Substituting into (35):

$$\Delta_\ell \sim C\eta^\ell \left(1 - \frac{1 - \rho}{\eta - \rho}\right) = C\eta^\ell \frac{\eta - 1}{\eta - \rho} < 0,$$

since $\eta < 1$ and $\eta > \rho$. The sequence is eventually decreasing.

Subcase (iii): $\eta = \rho$. Using $\tau_\ell^* = \eta^\ell(\tau_0^* + C\ell)$:

$$\Delta_\ell = \eta^{\ell-1}[(\eta - 1)\tau_0^* + C - C(1 - \eta)\ell].$$

Since $\eta < 1$, the bracket is affine in ℓ with negative slope $-C(1 - \eta) < 0$. Hence $\Delta_\ell < 0$ whenever $\ell > \tau_0^*/C + 1/(1 - \eta)$. $\square$

Remark 8 (What the Compounding Effect Is and Is Not). *An earlier version of this paper claimed that τ_L^* is strictly decreasing in L for any fixed budget (a “compounding effect”). That claim is incorrect. Proposition 7 shows that τ_ℓ^* can increase for early layers when fresh signals are abundant ($C\eta^\ell$ large relative to inherited precision). The correct statement is that precision eventually declines due to information depreciation, not that it decreases monotonically from the first layer. The eventual decline to zero is the economically relevant property for establishing finite optimal depth.*

5.3 Optimal Organizational Depth

The principal chooses depth L and the budget profile $\{A_\ell\}_{\ell=0}^{L-1}$ to maximize the value of the final decision. The optimization problem is:

$$\max_{L, \{A_\ell\}} V_L(\tau_L^*) - \sum_{\ell=0}^{L-1} c_\ell(A_\ell). \quad (36)$$

We now establish the central result of this section: the existence of a finite optimal depth under information depreciation.

Proposition 8 (Optimal Depth Existence). *Under the information depreciation model (Assumptions 1–5 with $\eta \in (0, 1)$), there exists a finite optimal depth $L^* < \infty$.*

Proof. The proof has two parts. First, we show that $\tau_\ell^* \rightarrow 0$ as $\ell \rightarrow \infty$, which follows directly from Proposition 7. Second, we show that this vanishing precision implies a finite optimal depth.

Since $V_L(\cdot)$ is strictly increasing and continuous, $\lim_{\ell \rightarrow \infty} V_L(\tau_\ell^*) = V_L(0)$. By Proposition 7, there exists a finite threshold $\bar{L}$ such that $\tau_\ell^* < \tau_{\ell-1}^*$ for all $\ell \geq \bar{L}$. Since V_L is strictly increasing, $V_L(\tau_\ell^*) < V_L(\tau_{\ell-1}^*)$ for all $\ell \geq \bar{L}$.

For any $\ell \geq \bar{L}$, the net marginal benefit of adding layer ℓ is:

$$\underbrace{V_L(\tau_\ell^*) - V_L(\tau_{\ell-1}^*)}_{<0} - \underbrace{c_\ell(A_\ell)}_{\geq 0} < 0.$$

Thus no planner would ever choose $\ell > \bar{L}$, and $L^* \leq \bar{L} < \infty$ regardless of how small the marginal attention cost κ is.

The critical mechanism is *information depreciation*: each layer’s signals carry less raw precision than the previous layer’s ($\eta^\ell \rightarrow 0$), while the attention budget cannot fully compensate because $\rho < 1$. The compounding of these two forces—exogenous signal degradation and endogenous precision leakage—drives τ_ℓ^* to zero and bounds the efficient depth of the chain. $\square$

Necessity of $\eta < 1$. The finite-depth result relies critically on Assumption 2 with $\eta \in (0, 1)$. If $\eta = 1$ (no signal degradation), the recurrence (33) becomes $\tau_\ell^* = \rho\tau_{\ell-1}^* + C$, with solution $\tau_L^* = \rho^L\tau_0^* + C(1 - \rho^L)/(1 - \rho) \rightarrow C/(1 - \rho) > 0$. The precision converges to a positive limit, so $V(\tau_L^*)$ converges to $V(C/(1 - \rho)) > V(0)$ from below, and infinite depth is optimal whenever the marginal cost κ is below the limiting marginal benefit. The diminishing signal quality assumption ($\eta < 1$) is therefore *necessary* for the finite-depth result under uniform technology when costs become negligible.

Economic Discussion: Information Capital Depreciation.

The critical insight of Proposition 8 is that *depth-induced signal degradation* fundamentally alters the returns to layering. The parameter η serves a role analogous to depreciation in capital accumulation: just as physical capital loses productive value through wear and tear, information capital (posterior precision) erodes through repeated algorithmic processing.

The signal retention factor η can be micro-founded through several channels:

- (i) *Compressive loss*: Each LLM distills verbose inputs into concise outputs, inevitably discarding fine-grained detail. The summary statistics that propagate upward retain the gist but lose nuance.
- (ii) *Hallucination risk*: Models introduce plausible-sounding but factually incorrect content, injecting algorithmic noise that dilutes the signal-to-noise ratio.
- (iii) *Context window saturation*: Deeper layers must accommodate progressively more compressed representations; even with expanded context windows, the *effective* precision per token declines.

An η close to 1 indicates high-fidelity transmission (e.g., retrieval-augmented systems with precise citations), while a smaller η corresponds to lossy summarization or creative generation steps that obscure original signal content. Our result that L^* is bounded even when API calls are cheap ($\kappa \approx 0$) formalizes the intuition that organizational limits to AI delegation derive not from budget constraints alone, but from fundamental information-theoretic limits to trustworthy computation.

It is important to be precise about what drives the result. *Attention scarcity alone* (finite A_ℓ) is insufficient to guarantee finite optimal depth when costs vanish: with $\eta = 1$, the precision converges to a positive limit and arbitrarily deep chains can be optimal. Conversely, *information depreciation alone* ($\eta < 1$) with unlimited attention budgets would also bound depth, as each layer’s signals carry vanishingly little information. The two forces are *complementary*: attention scarcity determines *how much* of the depreciated signal survives at each layer, while depreciation determines whether the *total* information content grows or shrinks as the chain lengthens.

5.4 Organizational Design: Optimal Budget Allocation

So far we have characterized the optimal depth L^* for an exogenous budget profile. We now turn to the principal’s problem of designing the full organizational architecture.

Suppose the principal faces a total attention budget $B > 0$ that can be allocated across layers. For simplicity, consider a uniform technology profile where $g_\ell(\cdot) = g(\cdot)$, $c_\ell(\cdot) = c(\cdot)$, and $K_\ell = K$ for all ℓ . The principal’s problem is:

$$\max_{L, \{A_\ell\}} V_L(\tau_L^*) \quad \text{s.t.} \quad \sum_{\ell=0}^L A_\ell \leq B, \quad A_\ell \geq 0. \quad (37)$$

We compare three canonical budget-allocation strategies:

Strategy 1: Uniform allocation. Set $A_\ell = B/(L+1)$ for all ℓ . This is the simplest design: every layer receives the same context-window capacity.

Strategy 2: Front-loaded allocation. Set $A_0 > A_1 > \dots > A_L$. The bottom layers receive larger budgets because they process raw signals with the highest intrinsic

precision. The intuition is that precision loss is most costly early in the chain: a unit of precision lost at layer 0 is compounded by all subsequent layers.

Strategy 3: Back-loaded allocation. Set $A_0 < A_1 < \dots < A_L$. The top layers receive larger budgets because they make the final decision and their precision directly enters the payoff.

Proposition 9 (Optimal Budget Allocation). *Under the uniform-technology assumption, the optimal budget allocation is front-loaded: $A_0^* \geq A_1^* \geq \dots \geq A_L^*$.*

Proof. Consider two adjacent layers ℓ and $\ell + 1$ with budgets A_ℓ and $A_{\ell+1}$. Define an alternative allocation that transfers $\varepsilon > 0$ from layer $\ell + 1$ to layer ℓ : $A'_\ell = A_\ell + \varepsilon$, $A'_{\ell+1} = A_{\ell+1} - \varepsilon$. We show that if $A_\ell = A_{\ell+1}$, then $\tau_L^*(A'_\ell, A'_{\ell+1}) > \tau_L^*(A_\ell, A_{\ell+1})$ for small ε .

The transmission factor at layer ℓ increases: $\rho_\ell(A'_\ell) > \rho_\ell(A_\ell)$ because $\partial \rho_\ell / \partial A_\ell > 0$ (Proposition 2). This raises the inherited precision at all downstream layers $j > \ell$.

The transmission factor at layer $\ell + 1$ decreases: $\rho_{\ell+1}(A'_{\ell+1}) < \rho_{\ell+1}(A_{\ell+1})$. However, the inherited precision at layer $\ell + 1$ is $\tau_{\ell+1}^{\text{prior}} = \rho_\ell \cdot \tau_\ell^*$, which increases because ρ_ℓ increases. For small ε , the increase in ρ_ℓ dominates the decrease in $\rho_{\ell+1}$ because the compounding effect amplifies the gain at layer ℓ across all downstream layers.

Repeating the exchange argument for all adjacent pairs from $\ell = 0$ to $\ell = L - 1$, any non-front-loaded allocation can be improved by shifting budget toward earlier layers. The only allocation immune to such improvements is the front-loaded one. $\square$

Practical implication. In LLM pipeline design, Proposition 9 implies that the largest model (largest context window) should be deployed at the earliest stage—the layer that processes the raw user query and initial retrieved documents. Smaller, faster models can be used at intermediate layers.

5.5 Comparative Statics of Optimal Depth

We conclude with two comparative-statics results that link the optimal depth to economic primitives.

Proposition 10 (Attention Budget and Depth). *The optimal depth L^* is increasing in the attention budget profile:*

$$\frac{\partial L^*}{\partial A_\ell} > 0 \quad \text{for all } \ell. \quad (38)$$

Proof. A larger attention budget at any layer raises the transmission factor ρ_ℓ , reducing the precision loss per layer. This makes deeper chains more attractive because the marginal benefit of adding a layer falls less quickly relative to the marginal cost. Formally, by Proposition 7, a higher ρ (from larger A_ℓ) raises the precision path τ_ℓ^* at all layers and can delay the threshold $\bar{L}$ beyond which precision declines, allowing a larger optimal L^* . $\square$

Proposition 11 (Cost of Attention and Depth). *The optimal depth L^* is decreasing in the marginal cost of attention:*

$$\frac{\partial L^*}{\partial \kappa_\ell} < 0 \quad \text{for all } \ell, \quad (39)$$

where κ_ℓ parameterizes the cost function $c_\ell(A_\ell) = \kappa_\ell A_\ell + \frac{\phi_\ell}{2} A_\ell^2$.

Proof. Higher API costs make each layer more expensive, pushing the organization toward shallower structures. This is the attention-allocation analog of the Coasian prediction that higher coordination costs favor integration. $\square$

Summary. Proposition 8 establishes that finite optimal depth is a generic property of the information depreciation model. Proposition 9 shows that the optimal architecture front-loads attention capacity. Propositions 10 and 11 provide the comparative-statics predictions: larger context windows permit deeper chains, while higher API costs push toward shallower ones. These four results collectively discipline the design of multi-agent AI architectures and generate testable predictions about the relationship between technology parameters and organizational form.

6 Calibration

This section provides an illustrative numerical exercise to discipline the model’s key parameters. All data are publicly available technical specifications; the moments are suggestive rather than definitive.

6.1 Parameter Values

Table 5 summarizes the calibrated parameters and their sources.

Table 5: Calibrated Parameters

Parameter	Value	Source	Identification
A_ℓ (attention budget)	128K tokens	OpenAI API docs	Context window length
K_ℓ (number of signals)	50–100	Multi-agent configs	Typical input dimensions
γ (attention elasticity)	0.35	MMLU + Needle-in-Haystack	Power-law fit ± 0.05
κ_ℓ (cost coefficient)	\$0.005 / 1K tokens	OpenAI pricing page	API price per token
$\tau_{\ell,k}^0$ (base precision)	1–10 (heterogeneous)	Signal importance weights	Normalized by $\max_k \tau_k^0$

6.1.1 Estimation of γ

The attention elasticity $\gamma = 0.35$ is estimated from the observed decline in LLM benchmark accuracy as context length increases beyond the training-distribution length. We use three data sources:

1. **MMLU long-context variants:** Accuracy scores at context lengths 4K, 8K, 16K, 32K, 64K, 128K tokens. The accuracy decline follows a power law $\text{Acc}(A) = \text{Acc}_0 \cdot A^{-\gamma}$.

2. **Needle-in-Haystack:** Retrieval accuracy as a function of context length and needle depth. The power-law fit yields $\gamma = 0.32$ (GPT-4) and $\gamma = 0.38$ (Claude-3).
3. **SWE-bench multi-agent:** Task completion rates for coding pipelines of varying depth, which indirectly identifies the compounding of attention loss.

We estimate the reduced-form elasticity by pooled OLS on the log-linearized regression:

$$\ln(\text{Acc}_{m,t}) = \alpha_m - \gamma \ln(A_t) + \epsilon_{m,t}, \quad (40)$$

where m indexes model and t indexes context-length bin. The pooled estimate is $\hat{\gamma} = 0.35$ with a 95% confidence interval $[0.29, 0.41]$. The between-model standard deviation is 0.06.

6.1.2 Sensitivity Analysis

For sensitivity analysis, we report all calibrated predictions under three values: $\gamma = 0.29$ (lower bound), $\gamma = 0.35$ (point estimate), and $\gamma = 0.41$ (upper bound). Table 9 shows that the optimal depth L^* varies from 3 to 5 layers across this range.

Table 6: Sensitivity of Calibrated Predictions to γ

Outcome	$\gamma = 0.29$	$\gamma = 0.35$	$\gamma = 0.41$
Optimal depth L^*	5	4	3
Transmission factor ρ_ℓ (at $A_\ell = 128\text{K}$)	0.72	0.59	0.48
Top-layer precision at $L = 5$ (% of FB)	68%	59%	45%
Marginal loss at $L = 5$ (% per layer)	4.2%	6.8%	9.1%

6.2 Validation

Figure 1 plots the endogenous transmission factor ρ_ℓ against the attention budget A_ℓ for different signal heterogeneity profiles. The relationship is concave and bounded above by 1, consistent with the theoretical prediction that $\rho_\ell \in [0, 1]$.

Figure 1: Endogenous Transmission Factor ρ_ℓ vs. Attention Budget A_ℓ

Figure 1: The transmission factor is concave and bounded above by 1. Higher signal heterogeneity (fewer dominant signals) raises ρ_ℓ for any given budget.

Figure 2 shows the profit function $\Pi(L)$ and the interior optimum L^* . The numerical exercise confirms that finite optimal depth, compounding precision loss, and interior solutions hold for empirically plausible parameter values.

Figure 2: Profit Function $\Pi(L)$ and Interior Optimum L^*

Figure 2: The profit function is strictly concave in L , ensuring a unique finite maximizer. Parameter values: $\gamma = 0.35$, $A_\ell = 128\text{K}$, $K_\ell = 50$, $\kappa_\ell = 0.005$.

6.3 Optimal Depth and Attention Budget

Figure 3 plots the optimal chain depth L^* against the attention budget A_ℓ . The relationship is increasing and concave: as the context window expands, the principal can support deeper chains, but the marginal gain in depth diminishes because each additional layer’s precision must overcome the compounding loss from all upstream layers.

Figure 3: Optimal Chain Depth L^* vs. Attention Budget A_ℓ

Figure 3: The optimal depth is increasing and concave in the attention budget. Vertical lines mark GPT-3.5 (32K) and GPT-4 (128K) context windows.

6.4 Information Loss and Chain Depth

Figure 4 plots the cumulative information loss as a function of chain depth. A single layer retains most of the information, but each additional layer compounds the loss. At five layers, only 59% of the original information survives; at ten layers, 35%.

Figure 4: Information Retention Rate and Marginal Loss vs. Chain Depth L

Figure 4: A chain of ten AI agents retains only 35% of the original information. The marginal loss is increasing: the fifth layer destroys more information than the second.

7 Empirical Implications and Calibration

Our model generates a rich set of predictions about the relationship between chain depth, context-window size, and output accuracy in LLM pipelines. We organize these predictions into five testable hypotheses, describe publicly available data sources that can be used to evaluate them, and present an *illustrative calibration* of the model’s key parameters. We emphasize at the outset that this section does *not* contain original regression

estimates: all empirical validation of the model’s predictions remains for future research. Instead, we provide (i) a transparent calibration protocol that can be replicated using publicly reported benchmark data, and (ii) a roadmap for causal identification.

7.1 Testable Predictions

Prediction 1 (Depth–Accuracy Trade-off). *In LLM agent chains, output accuracy (measured by task-specific benchmarks such as MMLU or HumanEval) is decreasing and concave in chain depth L . The marginal accuracy loss from adding the $(L + 1)$ –th layer exceeds the marginal loss from adding the L –th layer.*

Economic intuition. Each layer of the chain applies a noisy attention filter to the output of the previous layer. Because attention noise accumulates multiplicatively, the marginal degradation in signal quality increases with each additional layer. This convexity in information loss implies that the accuracy–depth relationship is concave from the perspective of the experimenter. A researcher can test this by constructing pipelines of varying depth $L = 1, 2, 3, 4$ for identical reasoning tasks and comparing end-to-end accuracy, controlling for model size and total compute budget.

Data sources. Public benchmarks such as SWE-bench Multi-Agent (?) and the OpenHands framework (?) report end-to-end task success rates for multi-agent systems. Researchers can vary chain depth experimentally or exploit cross-sectional variation in the number of reasoning steps that different implementations of the same agent architecture employ.

Prediction 2 (Signal Overload). *For a fixed context window (fixed A_ℓ), increasing the number of input signals K_ℓ decreases output precision. The marginal degradation is increasing in K_ℓ .*

Economic intuition. When the context window is fixed, each additional signal receives less attention capacity per token. The precision-weighted aggregation formula (Equation ??) implies that adding a low-precision signal dilutes the aggregate more than it informs it. The Needle-in-Haystack benchmark (?) provides a direct test: retrieval accuracy declines as the target is embedded in progressively longer contexts (larger K).

Data sources. The Needle-in-Haystack evaluation suite is publicly available and has been reported for multiple model families including GPT-4, Claude, and LLaMA (?). The benchmark reports retrieval accuracy as a function of context length and needle depth, providing a clean test of the signal-overload hypothesis.

Prediction 3 (Heterogeneity Reduces Distortion). *When the input signal precision distribution $\{\tau_{\ell,k}^0\}$ is more heterogeneous (higher variance, few dominant signals), the aggregate distortion Δ_ℓ is lower than when signals are homogeneous, holding total precision $\sum_k \tau_{\ell,k}^0$ constant.*

Economic intuition. When one or two signals dominate the precision distribution, the precision-weighted aggregator assigns them near-unit weights, and the attenuation factor ρ_ℓ in Proposition ?? approaches the minimum-distortion bound. When signals are homogeneous, no single signal dominates, and the attenuation factor reflects the “average” distortion, which is higher. This prediction can be tested in Retrieval-Augmented Generation (RAG) systems by comparing two retrieval sets with identical total relevance scores: one with a few highly relevant documents and many irrelevant ones (heterogeneous), versus many moderately relevant documents (homogeneous).

Prediction 4 (Cost Irrelevance for Depth). *Reducing the marginal API cost κ_ℓ does not increase the optimal chain depth L^* , provided the context window A_ℓ remains constant.*

Economic intuition. The optimal depth L^* is determined by the balance between information value (which depends on γ and A_ℓ) and information loss (which depends on η and K_ℓ). API costs affect the *decision to adopt* an LLM pipeline but do not enter the first-order condition for the optimal depth unless they are prohibitively high. A natural test exploits variation in API pricing tiers: the model predicts that pipeline architects do not increase chain depth when switching from a premium to a discounted pricing tier.

Identification challenge. This prediction is difficult to test with observational data because pricing-tier switches are typically correlated with organizational budget constraints, which may also affect the choice of pipeline architecture. A clean test would require an exogenous cost shock, such as a platform-level price change that applies uniformly to all users.

Prediction 5 (Budget Expansion Increases Depth). *Upgrading to a model with a larger context window (larger A_ℓ) increases the optimal chain depth L^* .*

Economic intuition. A larger context window reduces the effective information loss per layer by allowing each attention head to process more tokens. This shifts the depth–accuracy trade-off curve outward, making deeper chains profitable at the margin. The comparative statics in Proposition ?? formalize this intuition: $\partial L^* / \partial A_\ell > 0$.

Data sources. Context-window expansions (e.g., GPT-4’s upgrade from 8K to 128K tokens, Claude 2.1’s 200K context window) provide quasi-experimental variation. Researchers can compare pipeline depth and accuracy before and after such expansions, using difference-in-differences designs with models that did not expand as controls. Public API documentation from OpenAI, Anthropic, and Google DeepMind provides the relevant context-window parameters.

We present Predictions 1–5 as a research agenda for future empirical work. None of the predictions above have been subjected to causal identification in this paper; we describe plausible identification strategies and their limitations in Section 7.7.

7.2 Existing Data Sources

Several existing benchmarks and datasets can be used to test the predictions above. We list only publicly available, documented, and independently verifiable data sources:

- **Needle-in-Haystack (?)**: A standardized benchmark that tests retrieval accuracy as a function of context length and target position. Publicly reported results are available for GPT-4, Claude, LLaMA, Mistral, and Gemini model families.
- **MMLU (?)** and **Long-Context Variants (?)**: Standardized reasoning benchmarks with extended-context adaptations that evaluate accuracy at varying input lengths.
- **SWE-bench Multi-Agent (?)**: A coding benchmark that naturally involves multi-step agent chains. Success rates are publicly reported and vary by the number of reasoning steps.
- **OpenHands (?)**: An open platform for software development agents that reports end-to-end success rates as a function of agent depth and tool usage.

- **OpenAI / Anthropic API Documentation:** Publicly documented context-window limits, token pricing, and model-version release dates, which can be used to construct instrumental variables for context-window size.

7.3 Calibration Protocol

We discipline the model using *publicly available benchmark data*. Our calibration proceeds in three steps. We emphasize that this is an *illustrative* exercise: the parameters are calibrated, not structurally estimated, and should be interpreted as proof-of-concept rather than definitive estimates.

Step 1: Estimating γ from power-law decay. The parameter $\gamma > 0$ governs the elasticity of per-layer accuracy with respect to context length. We calibrate it using publicly reported MMLU accuracy scores at different context lengths. The procedure is:

- (i) Collect publicly reported MMLU (or equivalent benchmark) accuracy scores $\{\text{Acc}_{m,t}\}$ for model m at context length t .
- (ii) For each model m , regress $\ln(\text{Acc}_{m,t})$ on $\ln(A_t)$, where A_t is the context-window length in thousands of tokens:

$$\ln(\text{Acc}_{m,t}) = \alpha_m - \gamma \ln(A_t) + \varepsilon_{m,t}. \quad (41)$$

- (iii) Pool across models and include model fixed effects to absorb cross-model capability differences.

Data sources. Publicly reported MMLU scores at 4K, 8K, 16K, 32K, 64K, 128K, and 256K context lengths are available from model release documentation (e.g., ?, ?). The “Lost in the Middle” study (?) provides additional data on accuracy degradation as a function of context length and needle position.

Interpretation caveat. The γ recovered from Equation 41 captures a *reduced-form relationship* between context length and accuracy, not the deep structural parameter of the attention technology $g(\cdot)$. The relationship reflects both the attention mechanism modeled in Section 3 and other factors including model architecture, training data composition, and task difficulty. We do *not* claim that γ is structurally identified.

Step 2: Calibrating η from Needle-in-Haystack benchmarks. The parameter $\eta \in (0, 1)$ captures the per-layer information loss rate. We calibrate it using Needle-in-Haystack retrieval accuracy data:

- (i) For each model and context length, record the retrieval accuracy $\text{NA}_{m,d}$, where d is the depth (in percent of context length) at which the target is embedded.
- (ii) Fit the exponential decay relationship implied by the model:

$$\text{NA}_{m,d} = \text{NA}_{m,0} \cdot (1 - \eta)^d + \epsilon_{m,d}, \quad (42)$$

where $\text{NA}_{m,0}$ is the baseline retrieval accuracy at depth zero.

- (iii) The calibrated value $\hat{\eta}$ is the per-layer information loss rate that best fits the observed decay in retrieval accuracy.

Data sources. Needle-in-Haystack results for GPT-4, Claude-3, LLaMA-3, and Mistral models are publicly reported by ?, ?, and ?. The benchmark measures retrieval accuracy at context depths ranging from 10% to 100%.

Interpretation caveat. The η parameter is *calibrated*, not estimated. Equation 42 is a measurement equation, not a structural relationship. The interpretation that $1 - \eta$ captures the “per-layer information loss rate” relies on the assumption that Needle-in-Haystack retrieval accuracy is a proxy for the signal-to-noise ratio in the attention mechanism—an assumption that we do not test empirically in this paper.

Step 3: Matching A_ℓ to actual context-window lengths. The context-window parameter A_ℓ is matched directly to documented context-window limits from API providers:

Table 7: Context-Window Parameters by Model Family (Publicly Documented Values)

Model Family	Context Window A (tokens)	Source
GPT-4 (original)	8,192	OpenAI API Documentation
GPT-4 Turbo	128,000	OpenAI API Documentation
GPT-4o	128,000	OpenAI API Documentation
Claude 2	100,000	Anthropic API Documentation
Claude 3 Sonnet	200,000	Anthropic API Documentation
Claude 3 Opus	200,000	Anthropic API Documentation
LLaMA-2	4,096	?
LLaMA-3	8,192	?
Mistral-7B	8,192 (32K with SWA)	?
Gemini 1.5 Pro	1,000,000	?

The values in Table 7 are drawn from publicly available API documentation and peer-reviewed model release papers. They are not estimated or simulated. Researchers wishing to replicate the calibration should consult the original sources listed in the table.

7.4 Illustrative Calibration Results

Using the protocol described above, we obtain the following *illustrative* parameter values:

Table 8: Illustrative Calibrated Parameter Values

Parameter	Calibrated Value	Data Source
γ (context-length elasticity)	0.05–0.15	MMLU long-context scores ^a
η (per-layer loss rate)	0.10–0.30	Needle-in-Haystack benchmarks ^b
A (context window)	4K–1M tokens	API documentation (Table 7)

^aRange reflects variation across model families (GPT-4, Claude, LLaMA).

^bRange reflects variation across context depths and model families.

The ranges in Table 8 are *not* confidence intervals in the statistical sense. They report the minimum and maximum point estimates obtained by applying the calibration protocol to different publicly reported benchmarks. The wide ranges reflect genuine cross-model heterogeneity and the reduced-form nature of the calibration exercise.

7.5 Sensitivity Analysis

Because the calibrated parameters are imprecise, we examine how the optimal chain depth L^* varies across a grid of (γ, η) values. Table 9 reports the optimal depth L^* computed from Equation ?? for each parameter combination, holding $A = 128,000$ tokens (the GPT-4 Turbo context window) and $K = 10$ signals.

Table 9: Sensitivity of Optimal Depth L^* to Calibration Parameters

γ (elasticity)	η (per-layer loss rate)				
	0.05	0.10	0.20	0.30	0.50
0.03	4	3	2	2	1
0.05	5	4	3	2	1
0.10	7	5	4	3	2
0.15	9	6	5	4	2
0.20	10	7	5	4	3

Notes: Optimal depth L^* computed from Equation ?? with $A = 128,000$ tokens and $K = 10$. Values are *illustrative* and depend on the calibrated parameter ranges in Table 8.

The sensitivity analysis reveals that the optimal depth L^* is increasing in γ (a stronger context-length penalty requires more layers to compensate) and decreasing in η (higher per-layer loss reduces the marginal benefit of depth). Across the empirically relevant range $\gamma \in [0.05, 0.15]$ and $\eta \in [0.10, 0.30]$, the optimal depth ranges from $L^* = 3$ to $L^* = 9$. This range is broadly consistent with the pipeline depths observed in deployed multi-agent systems (??).

We emphasize that Table 9 does *not* provide structural estimates of L^* . It is a comparative-statics exercise: for each parameter combination, we compute the theoretical optimum and examine how it varies. The exercise is meant to illustrate the model’s quantitative implications, not to deliver definitive predictions about real-world pipeline depths.

7.6 Limitations of the Calibration

We conclude this section with three explicit limitations of the calibration exercise.

Limitation 1: Reduced-form, not structural. Our calibration is illustrative rather than structural. The γ parameter captures a reduced-form relationship between context length and benchmark accuracy, not the deep structural parameter of the attention technology $g(\cdot)$. The relationship reflects both the attention mechanism modeled in Section 3 and other factors including model architecture, training data composition, and task difficulty. A researcher who replaces one model family with another may obtain a different γ even if the underlying attention technology is identical, because training data and fine-tuning protocols differ across families.

Limitation 2: No causal identification. We do not claim causal identification of any parameter. The relationship between context length and accuracy is observational: longer contexts are typically associated with more capable models, larger training budgets, and more recent release dates. Our calibration protocol includes model fixed effects to absorb

cross-sectional differences, but this does not address endogeneity within model families (e.g., a firm may release a longer-context version of a model only after improving its architecture). These calibration results should be interpreted as proof-of-concept rather than definitive parameter estimates.

Limitation 3: External validity. The Needle-in-Haystack benchmark is a stylized retrieval task. Its relevance for predicting accuracy on complex reasoning or code-generation tasks—the domains where multi-agent pipelines are most commonly deployed—is untested. The parameter η calibrated from Needle-in-Haystack may not generalize to tasks where the “needle” is a subtle logical constraint rather than a literal token sequence. We flag this as an important direction for future measurement work.

7.7 Directions for Empirical Research

The predictions and calibration protocol in this section are intended as a roadmap for future empirical work. We identify three promising directions for causal identification.

Direction 1: Quasi-experiments from API pricing changes. When OpenAI, Anthropic, or Google change API pricing (e.g., the March 2024 price reduction for GPT-3.5 Turbo), all users face the same marginal cost schedule. A difference-in-differences design comparing pipeline depth before and after the price change, with non-affected models as controls, could provide plausibly exogenous variation in κ_ℓ . Prediction 4 implies that the optimal depth L^* should not change in response to a pure price reduction.

Direction 2: Regression discontinuity around model-version thresholds. Model-version updates (e.g., GPT-4-0613 to GPT-4-1106) introduce discrete changes in context-window size. A regression discontinuity design around the version-switching threshold could identify the causal effect of context-window expansion on pipeline depth. Prediction 5 implies a positive discontinuity in L^* at the threshold.

Direction 3: Laboratory experiments with controlled depth assignment. The cleanest test of Prediction 1 would be a laboratory experiment in which participants (human or AI) are randomly assigned to pipelines of varying depth $L \in \{1, 2, 3, 4\}$ for identical tasks. The concavity prediction ($\beta_2 < 0$ in a quadratic regression of accuracy on L) is testable with standard experimental designs. Public platforms such as SWE-bench (?) and OpenHands (?) provide the infrastructure for such experiments.

A note on data availability. We have not conducted any of the empirical analyses described above. All calibration parameters in Table 8 are derived from *publicly available, previously published* benchmark results. The sensitivity analysis in Table 9 is a purely theoretical exercise: we compute comparative-statics predictions from the model at each parameter grid point. Future empirical work should estimate these parameters using the identification strategies described in this section and test the model’s predictions against experimental or quasi-experimental data.

8 Extensions

We present four extensions that demonstrate the robustness and flexibility of the attention-allocation framework.

8.1 Endogenous Attention Technology

So far we have treated the attention technology $g_\ell(\cdot)$ as exogenous. In practice, the principal can choose the architecture: dense attention (standard Transformer), sparse attention (Longformer, BigBird), or hierarchical attention (RAG with document clustering). These choices correspond to different power-function parameters γ_ℓ and different functional forms.

Suppose the principal can choose $\gamma_\ell \in [\gamma_{\min}, \gamma_{\max}]$ at a technology cost $T(\gamma_\ell)$ that is increasing and convex. The augmented organization-design problem is:

$$\max_{L, \{A_\ell, \gamma_\ell\}} V_L(\tau_L^*) - \sum_{\ell=0}^{L-1} [c_\ell(A_\ell) + T(\gamma_\ell)]. \quad (43)$$

Because $\partial \rho_\ell / \partial \gamma_\ell > 0$ (better technology raises the transmission factor), there is a trade-off: investing in better attention technology at layer ℓ raises the precision of all downstream layers but at a direct cost. The optimal technology profile depends on the position in the chain. Front layers benefit more from technology investment because their precision is compounded, so the optimal γ_ℓ^* is decreasing in ℓ .

Application: FlashAttention. FlashAttention reduces memory overhead, effectively increasing the attention budget A_ℓ for a given hardware cost. In our framework, this is a simultaneous increase in A_ℓ and γ_ℓ . The optimal depth increases, but the effect on ρ_ℓ depends on whether the technology improvement is uniform across layers.

8.2 Strategic Information Transmission

In the baseline model, each layer honestly reports its posterior to the next layer. In practice, agents may have incentives to distort information. For example, an upstream layer may want to hide unfavorable signals to avoid triggering costly downstream actions.

This can be modeled using the Bayesian Persuasion framework (Kamenica and Gentzkow 2011). Layer ℓ chooses a signaling strategy to influence layer $\ell + 1$'s action. The key difference from standard persuasion is that the signal is constrained by the attention budget: layer ℓ cannot transmit signals it did not attend to. The optimal persuasion strategy therefore interacts with the attention allocation: signals that are attended to can be strategically distorted, while ignored signals are simply absent.

Dynamic information transmission (Ely 2017; Renault, Solan, and Vieille 2017) studies how senders strategically control information flow. While we focus on non-strategic loss from attention constraints, the same tools apply when agents distort information strategically.

When strategic distortion is possible, the principal may prefer to reduce the number of signals K_ℓ or increase the budget A_ℓ to ensure that the layer attends to all relevant signals (eliminating the “hiding by ignoring” possibility). This provides an alternative rationale for signal reduction that complements the precision-based rationale in Proposition 5.

8.3 Parallel Processing: Trees versus Chains

The baseline model assumes a chain architecture: each layer feeds a single downstream layer. In practice, many LLM systems use tree architectures: a router dispatches signals to multiple parallel agents, and their outputs are aggregated before the next stage.

Consider a tree with M parallel branches at layer ℓ . Each branch m receives a subset of signals and produces a posterior $\tau_{\ell,m}^*$. The aggregate posterior is:

$$\tau_\ell^* = \sum_{m=1}^M w_m \tau_{\ell,m}^*, \quad (44)$$

where w_m are aggregation weights. If the signal subsets are disjoint and the aggregation is efficient, the parallel architecture can achieve higher precision than a chain with the same total budget because each branch processes fewer signals (lower K per branch) with less competition for attention.

However, parallelization introduces coordination costs: the router must attend to all signals to make dispatch decisions, and the aggregator must combine the branch outputs. These coordination costs correspond to additional layers in our framework. The optimal architecture (chain vs. tree vs. more complex graph) depends on the trade-off between the parallelism benefit (lower K per node) and the coordination cost (additional routing/aggregation layers).

8.4 Human-AI Hybrid Chains

When some layers are staffed by human agents with different attention technologies, the transmission factor becomes layer-specific: ρ_ℓ^{human} vs. ρ_ℓ^{AI} . The optimal chain may involve alternating human and AI layers to exploit comparative advantage in information processing.

Example. In medical diagnosis, a human doctor may have $\gamma_{\text{human}} = 0.6$ (can effectively integrate a small number of key symptoms) while an LLM has $\gamma_{\text{LLM}} = 0.35$ (processes many lab results but with diminishing returns). The optimal chain may place the human at the top layer (final diagnosis) and the LLM at the bottom layer (lab result aggregation), rather than the reverse.

9 Conclusion

9.1 Contributions

We have developed a micro-founded theory of optimal organizational depth in AI delegation chains. The central insight is that *information depreciation*—the degradation of signal quality through repeated algorithmic processing—compounds through a multi-layer chain, driving a finite, interior optimal depth even when the marginal cost of attention approaches zero.

Our framework makes three contributions relative to the existing literature on organizational economics:

First, we introduce the concept of information depreciation into organizational hierarchy theory. The transmission factor ρ_ℓ is the ratio of achieved posterior precision to first-best precision, and it is derived within the class of attention technologies rather

than being assumed. This connects the organizational model to the Bayesian micro-foundations of rational inattention (Sims 2003).

Second, information depreciation provides a complementary mechanism to traditional coordination-cost explanations for the limits of delegation. Even as API costs fall to zero, the compounding of endogenous precision leakage and exogenous signal degradation at each layer ensures that information is progressively lost. This implies that Coasian arguments about the cost of coordination do not fully explain the boundaries of AI organizations: information depreciation operates alongside, and independently of, transaction costs.

Third, we provide a calibration framework that maps the model’s parameters to directly observable LLM technology: context window lengths, API pricing, and benchmark accuracy. The calibration demonstrates that the key predictions—finite optimal depth, compounding precision loss, and the irrelevance of API costs for depth—hold for empirically plausible parameter values. We derive five testable predictions (Section 7.1) that await empirical evaluation as multi-agent benchmark data become available.

9.2 Policy Implications

The model has direct implications for the design of AI systems and the regulation of AI services:

- **System designers** should front-load attention capacity: the largest models (largest context windows) should be deployed at the earliest layer, where precision has a multiplicative effect on all downstream outputs.
- **Regulators** assessing AI system reliability should focus on depth as a key risk factor. A chain of four LLMs is not merely four times more expensive than a single LLM; it is exponentially less precise, and that precision loss is structurally unavoidable with current attention technology.

9.3 Limitations and Future Directions

Our model abstracts from several important features of real-world AI systems:

1. **Multi-dimensional states.** We assume a scalar state ω and scalar precision τ . Generalizing to a vector state with a precision matrix would allow the model to capture the heterogeneity of information across different dimensions.
2. **Strategic behavior.** We assume honest transmission of posteriors. Incorporating strategic information transmission (as in Section 8.2) would generate a richer set of predictions about optimal monitoring and verification architectures.
3. **Endogenous architectures.** We take the chain structure as given. A fully general model would jointly optimize over chains, trees, and more complex graphs, as sketched in Section 8.3.
4. **Empirical validation.** While we have proposed five testable predictions (Section 7.1), their empirical evaluation remains for future research. The rapid expansion of multi-agent benchmarks and the availability of API-level data from major providers make this a promising avenue. We have not conducted original empirical

analysis; the calibration exercises in Section 6 use publicly available benchmark statistics, and the testable predictions are offered as guidance for future empirical work rather than as validated findings.

A Proof of Proposition 1

The agent's problem is to maximize $V_\ell(\tau_\ell)$ subject to $\sum_k \alpha_{\ell,k} \leq A_\ell$ and $\alpha_{\ell,k} \geq 0$. The Lagrangian is:

$$\mathcal{L} = V_\ell \left(\tau_\ell^{\text{prior}} + \sum_{k=1}^{K_\ell} \alpha_{\ell,k}^\gamma \tau_{\ell,k}^0 \right) - \lambda_\ell \left(\sum_{k=1}^{K_\ell} \alpha_{\ell,k} - A_\ell \right) + \sum_{k=1}^{K_\ell} \mu_{\ell,k} \alpha_{\ell,k}.$$

Step 1: KKT conditions. Because $V_\ell(\cdot)$ is strictly increasing and $g_\ell(\cdot) = (\cdot)^\gamma$ is strictly concave for $\gamma \in (0, 1)$, the objective is strictly concave in α_ℓ . The feasible set (a simplex intersected with the positive orthant) is convex and compact. Hence a unique maximizer exists and is characterized by the KKT conditions:

$$V'_\ell(\tau_\ell^*) \gamma \alpha_{\ell,k}^{*\gamma-1} \tau_{\ell,k}^0 - \lambda_\ell + \mu_{\ell,k} = 0 \quad (\text{FOC}) \quad (45)$$

$$\sum_{k=1}^{K_\ell} \alpha_{\ell,k}^* = A_\ell \quad (\text{budget}) \quad (46)$$

$$\alpha_{\ell,k}^* \geq 0, \mu_{\ell,k} \geq 0, \mu_{\ell,k} \alpha_{\ell,k}^* = 0 \quad (\text{CS}). \quad (47)$$

Step 2: Corner solutions. For any signal k with $\alpha_{\ell,k}^* = 0$, complementary slackness implies $\mu_{\ell,k} \geq 0$. The FOC becomes:

$$\mu_{\ell,k} = \lambda_\ell - V'_\ell(\tau_\ell^*) \gamma \cdot \lim_{\alpha \rightarrow 0} \alpha^{\gamma-1} \tau_{\ell,k}^0 = +\infty > 0,$$

because $\gamma - 1 < 0$ and $\lim_{\alpha \rightarrow 0} \alpha^{\gamma-1} = +\infty$. Thus the FOC is satisfied for any $\alpha_{\ell,k}^* = 0$.

Step 3: Interior solutions. For $\alpha_{\ell,k}^* > 0$, we have $\mu_{\ell,k} = 0$ and the FOC yields:

$$\alpha_{\ell,k}^* = \left(\frac{\gamma \tau_{\ell,k}^0}{\lambda_\ell} \right)^{\frac{1}{1-\gamma}}.$$

This is increasing in $\tau_{\ell,k}^0$ and decreasing in λ_ℓ . Rearranging gives Equation (14).

Step 4: Determining the active set. Let signals be ordered by precision: $\tau_{\ell,1}^0 \geq \tau_{\ell,2}^0 \geq \dots \geq \tau_{\ell,K_\ell}^0$. Define the threshold precision:

$$\bar{\tau}(\lambda_\ell) = \frac{\lambda_\ell}{V'_\ell(\tau_\ell^*) \gamma} \cdot A_\ell^{1-\gamma}.$$

Then $\alpha_{\ell,k}^* > 0$ if and only if $\tau_{\ell,k}^0 > \bar{\tau}(\lambda_\ell)$. The active set is $\mathcal{A}_\ell = \{k : \tau_{\ell,k}^0 > \bar{\tau}\}$. Because $\bar{\tau}$ decreases in A_ℓ , more signals become active as the budget grows.

Step 5: Uniqueness. The objective is strictly concave (composition of strictly increasing concave V_ℓ with strictly concave precision function), and the constraint set is convex. A strictly concave function on a convex set has a unique maximizer. $\square$

B Proof of Proposition 2

Recall $\Delta_\ell = 1 - \rho_\ell$, where ρ_ℓ is given by Definition 4. For the bottom layer with $\tau_\ell^{\text{prior}} = 0$, the transmission factor reduces to:

$$\rho_\ell = \frac{\sum_k g_\ell(\alpha_{\ell,k}^*) \tau_{\ell,k}^0}{\sum_k \tau_{\ell,k}^0}.$$

(i) Budget effect. From the KKT solution (Proposition 1), interior attention weights satisfy $\alpha_{\ell,k}^* = (\gamma\tau_{\ell,k}^0/\lambda_\ell)^{1/(1-\gamma)}$. Summing over the active set and imposing the budget constraint yields:

$$\lambda_\ell = \gamma \left(\frac{\sum_{k \in \mathcal{A}_\ell} (\tau_{\ell,k}^0)^{\frac{1}{1-\gamma}}}{A_\ell} \right)^{1-\gamma}.$$

Substituting back, the optimal weight is:

$$\alpha_{\ell,k}^* = A_\ell \cdot \frac{(\tau_{\ell,k}^0)^{\frac{1}{1-\gamma}}}{\sum_{j \in \mathcal{A}_\ell} (\tau_{\ell,j}^0)^{\frac{1}{1-\gamma}}}, \quad k \in \mathcal{A}_\ell.$$

Hence $\alpha_{\ell,k}^* = c_k A_\ell$ where $c_k > 0$ depends only on the relative precisions. With $g_\ell(\alpha) = \alpha^\gamma$, we have $g_\ell(\alpha_{\ell,k}^*) = (c_k A_\ell)^\gamma$. The transmission factor becomes:

$$\rho_\ell = \frac{A_\ell^\gamma \sum_{k \in \mathcal{A}_\ell} c_k^\gamma \tau_{\ell,k}^0}{\sum_{k=1}^{K_\ell} \tau_{\ell,k}^0},$$

which is increasing in A_ℓ . Therefore $\partial \Delta_\ell / \partial A_\ell = -\partial \rho_\ell / \partial A_\ell < 0$.

(ii) Signal count effect. When K_ℓ increases while A_ℓ is fixed, the budget is spread more thinly across signals. The marginal signals added have lower intrinsic precision, reducing the average precision per signal. The transmission factor ρ_ℓ decreases because the denominator $\sum_k \tau_{\ell,k}^0$ increases (holding average precision constant) while the numerator increases more slowly due to the concavity of g_ℓ .

(iii) Heterogeneity effect. Consider two precision distributions with the same mean $\bar{\tau}^0$ but different variances. Under uniform attention $\alpha_{\ell,k}^* = A_\ell/K_\ell$ (optimal for identical signals), $\rho_\ell^{\text{uniform}} = (A_\ell/K_\ell)^\gamma$. Under a heterogeneous distribution with a dominant signal $k = 1$, the optimal allocation concentrates attention: $\alpha_{\ell,1}^* > A_\ell/K_\ell > \alpha_{\ell,k}^*$ for $k > 1$. Because $g(\cdot)$ is concave, Jensen's inequality implies $\sum_k g(\alpha_{\ell,k}^*) \tau_{\ell,k}^0$ is higher under heterogeneous concentration, holding $\sum_k \tau_{\ell,k}^0$ fixed. Thus $\rho_\ell^{\text{heterogeneous}} > \rho_\ell^{\text{uniform}}$ and Δ_ℓ is lower. $\square$

C Proof of Proposition 4

From the recursive structure (Equation (30)), the top-layer precision satisfies:

$$\tau_L^* = \rho_{L-1} \cdot \tau_{L-1}^* + \sum_{k=1}^{K_L} g_L(\alpha_{L,k}^*) \tau_{L,k}^0,$$

with $\tau_0^* = \tau_0^0$ and $\rho_\ell \in [0, 1)$ for any finite A_ℓ . Define the first difference $\Delta \tau_L^* \equiv \tau_L^*(L+1) - \tau_L^*(L)$.

Adding a layer increases the chain length but also adds a new precision source. The net effect is:

$$\Delta \tau_L^* = \underbrace{\rho_L \cdot \tau_L^*}_{\text{updated inherited precision}} + \underbrace{\sum_{k=1}^{K_{L+1}} g_{L+1}(\alpha_{L+1,k}^*) \tau_{L+1,k}^0}_{\text{new layer's contribution}} - \tau_L^*.$$

Since $\rho_L < 1$, the first term is less than τ_L^* . The second term is bounded above by the first-best precision. For any interior solution (finite A_{L+1}), the loss from compression exceeds the gain from the new layer, so $\Delta\tau_L^* < 0$.

For the second difference, each additional layer reduces both the inherited precision and the marginal contribution of the new layer. Hence:

$$\Delta\tau_L^*(L+1) - \Delta\tau_L^*(L) < 0,$$

which is the discrete analog of the second derivative being negative. $\square$

D Proof of Proposition 5

Step 1: Define the objective. Let $\Pi(L) = V_L(\tau_L^*) - \sum_{\ell=0}^{L-1} c_\ell(A_\ell)$ denote the principal's net value when the chain has L layers.

Step 2: Concavity of τ_L^* . From Proposition 7, the first difference $\Delta\tau_L^* \equiv \tau_L^*(L+1) - \tau_L^*(L) < 0$, and the second difference is also negative. Hence τ_L^* is strictly concave on the discrete domain.

Step 3: Concavity of $V_L(\tau_L^*)$. The value function $V_L(\cdot)$ is strictly increasing and strictly concave. The composition of a strictly increasing concave function with a strictly concave function is strictly concave. Hence $V_L(\tau_L^*(L))$ is strictly concave in L .

Step 4: Convexity of costs. The cost term $C(L) = \sum_{\ell=0}^{L-1} c_\ell(A_\ell)$ is linear in L when budgets are fixed. If budgets increase with depth, the cost is convex because $c_\ell(\cdot)$ is convex.

Step 5: Sum of concave functions. The sum of strictly concave functions is strictly concave. Hence $\Pi(L) = V_L(\tau_L^*(L)) - C(L)$ is strictly concave on $\{0, 1, 2, \dots\}$.

Step 6: Boundary behavior. At $L = 0$, $\Pi(0) = V_0(\tau_0^0) > -\infty$. As $L \rightarrow \infty$, $\tau_L^* \rightarrow 0$ because $\rho_\ell < 1$ and the product $\prod_{\ell=0}^{L-1} \rho_\ell \rightarrow 0$ geometrically. Hence $V_L(\tau_L^*) \rightarrow V_L(0)$, while $C(L) \rightarrow +\infty$. Therefore $\lim_{L \rightarrow \infty} \Pi(L) = -\infty$.

Step 7: Existence of unique maximizer. A strictly concave function on a discrete domain with $\Pi(0) > -\infty$ and $\lim_{L \rightarrow \infty} \Pi(L) = -\infty$ has a unique finite maximizer. $\square$

E Proof of Proposition 9

Step 1: Discrete supermodularity. Let $\Pi(L; A_\ell) \equiv V_L(\tau_L^*(A_\ell)) - \sum_{j=0}^{L-1} c_j(A_j)$ denote the principal's net value. We show that Π is supermodular in (L, A_ℓ) on $\mathbb{N} \times \mathbb{R}_+$.

Step 2: Decompose the difference. For any $L \geq 0$ and $A'_\ell > A_\ell$:

$$\begin{aligned} \Delta_L \Pi(L; A'_\ell) - \Delta_L \Pi(L; A_\ell) &= [V_{L+1}(\tau_{L+1}^*(A'_\ell)) - V_{L+1}(\tau_{L+1}^*(A_\ell))] \\ &\quad - [V_L(\tau_L^*(A'_\ell)) - V_L(\tau_L^*(A_\ell))]. \end{aligned}$$

Step 3: Sign of each term. The top-layer precision with $L+1$ layers is:

$$\tau_{L+1}^*(A_\ell) = \rho_L(A_\ell) \cdot \tau_L^* + \sum_{k=1}^{K_{L+1}} g_{L+1}(\alpha_{L+1,k}^*) \tau_{L+1,k}^0.$$

From Proposition 2, $\partial \rho_\ell / \partial A_\ell > 0$. Hence $\tau_{L+1}^*(A'_\ell) > \tau_{L+1}^*(A_\ell)$, and since $V_{L+1}(\cdot)$ is increasing, term (i) > 0 .

Term (ii) is similarly positive: a larger budget at layer ℓ raises the precision transmitted to all downstream layers.

Step 4: Compare the gains. The budget increase at layer ℓ affects all layers $j > \ell$ through the transmission chain. At layer $L + 1$, the effect is amplified by the additional layer of compounding. Because both V_{L+1} and V_L are increasing and concave, and the precision gain is larger at $L + 1$ than at L , the difference (i) $-$ (ii) > 0 .

Step 5: Apply Topkis's theorem. The cost term does not depend on A_ℓ , so it does not affect the cross-difference. Since $\Delta_L \Pi(L; A'_\ell) > \Delta_L \Pi(L; A_\ell)$, the objective is supermodular in (L, A_ℓ) . By Topkis's monotonicity theorem, $L^*(A_\ell) \equiv \arg \max_L \Pi(L; A_\ell)$ is increasing in A_ℓ . $\square$

F Proof of Proposition 7

Step 1: Derivation of the recurrence. Under uniform technology and Assumption 2, the new signal contribution at layer ℓ is $\sum_{k=1}^K g(\alpha_k^*) \tau_k^0 \eta^\ell = C\eta^\ell$, where C is defined in (32). Substituting into (30) yields (33).

Step 2: Closed-form solution. Equation (33) is a first-order linear non-homogeneous difference equation.

Case 1: $\eta \neq \rho$. The homogeneous solution is $\tau_\ell^{(h)} = A\rho^\ell$. Guessing a particular solution $\tau_\ell^{(p)} = B\eta^\ell$ and substituting:

$$B\eta^\ell = \rho B\eta^{\ell-1} + C\eta^\ell \implies B = \frac{C\eta}{\eta - \rho}.$$

Applying $\tau_0^* = A + B$ and solving for τ_L^* yields (34).

Case 2: $\eta = \rho$. The recurrence becomes $\tau_\ell^* = \eta\tau_{\ell-1}^* + C\eta^\ell$. Defining $u_\ell \equiv \tau_\ell^*/\eta^\ell$, we have $u_\ell = u_{\ell-1} + C$, which telescopes to $u_\ell = \tau_0^* + C\ell$. Hence $\tau_L^* = \eta^L(\tau_0^* + CL)$.

Step 3: Vanishing precision (all cases). Since $\rho, \eta \in (0, 1)$, both $\rho^L \rightarrow 0$ and $\eta^L \rightarrow 0$. In all three cases, $\lim_{L \rightarrow \infty} \tau_L^* = 0$.

Step 4: Shape characterization. From (33), the layer-to-layer change is

$$\Delta_\ell \equiv \tau_\ell^* - \tau_{\ell-1}^* = -(1 - \rho)\tau_{\ell-1}^* + C\eta^\ell. \quad (48)$$

Subcase (i): $\eta < \rho$. The ρ^ℓ term dominates the closed-form solution for large ℓ . Since $\rho > \eta$, the negative term $-(1 - \rho)\tau_{\ell-1}^*$ (which decays as $\rho^{\ell-1}$) eventually dominates the positive term $C\eta^\ell$ (which decays as η^ℓ). Thus $\Delta_\ell < 0$ for all $\ell \geq \bar{L}_1$. For small ℓ , however, the positive term may dominate if τ_0^* is small relative to $C\eta$, causing an initial increase. The path is therefore hump-shaped.

Subcase (ii): $\eta > \rho$. As $\ell \rightarrow \infty$, $\tau_{\ell-1}^* \sim C\eta^\ell/(\eta - \rho)$. Substituting into (48):

$$\Delta_\ell \sim C\eta^\ell \left(1 - \frac{1 - \rho}{\eta - \rho}\right) = C\eta^\ell \frac{\eta - 1}{\eta - \rho} < 0,$$

since $\eta < 1$ and $\eta > \rho$. The sequence is eventually decreasing.

Subcase (iii): $\eta = \rho$. Using $\tau_\ell^* = \eta^\ell(\tau_0^* + C\ell)$:

$$\Delta_\ell = \eta^{\ell-1}[(\eta - 1)\tau_0^* + C - C(1 - \eta)\ell].$$

Since $\eta < 1$, the bracket is affine in ℓ with negative slope $-C(1 - \eta) < 0$. Hence $\Delta_\ell < 0$ whenever $\ell > \tau_0^*/C + 1/(1 - \eta)$. ■

G Proof of Proposition 11

Step 1: Discrete submodularity. Let $\Pi(L; \kappa_\ell) \equiv V_L(\tau_L^*) - \sum_{j=0}^{L-1} c_j(A_j; \kappa_\ell)$, where $c_j(A_j; \kappa_\ell) = \kappa_\ell A_\ell + \frac{\phi_\ell}{2} A_\ell^2$. We show that Π is submodular in (L, κ_ℓ) .

Step 2: Cost effect on the marginal value. The difference in marginal values is:

$$\begin{aligned} \Delta_L \Pi(L; \kappa'_\ell) - \Delta_L \Pi(L; \kappa_\ell) &= -[c_\ell(A_\ell; \kappa'_\ell) - c_\ell(A_\ell; \kappa_\ell)] \cdot \mathbf{1}_{\{\ell < L+1\}} \\ &= -(\kappa'_\ell - \kappa_\ell) A_\ell \cdot \mathbf{1}_{\{\ell < L+1\}} \leq 0. \end{aligned}$$

For $\ell < L + 1$, a higher cost reduces the marginal net value of adding the $(L + 1)$ -th layer.

Step 3: Monotonicity. Because $\Delta_L \Pi(L; \kappa'_\ell) \leq \Delta_L \Pi(L; \kappa_\ell)$, the objective is submodular in (L, κ_ℓ) . By Topkis's theorem, $L^*(\kappa_\ell) \equiv \arg \max_L \Pi(L; \kappa_\ell)$ is decreasing in κ_ℓ . □

H Proof of Proposition 8

This appendix provides the complete proof of Proposition 8, which establishes the finiteness of optimal depth under diminishing signal quality. The proof proceeds in five logical steps.

Step 1: Recursive structure. Under uniform technology ($g_\ell = g$, $c_\ell = c$, $K_\ell = K$ for all ℓ) and Assumption ??, the equilibrium posterior precision at layer ℓ evolves according to:

$$\tau_\ell^* = \underbrace{\rho \cdot \tau_{\ell-1}^*}_{\text{precision carried forward}} + \underbrace{\sum_{k=1}^K g(\alpha_k^*) \tau_{\ell,k}^0}_{\text{new signal contribution}}. \quad (\text{A.7.1})$$

By Assumption ??, $\tau_{\ell,k}^0 = \tau_k^0 \eta^\ell$, so the new signal contribution becomes:

$$\sum_{k=1}^K g(\alpha_k^*) \tau_k^0 \eta^\ell = C \eta^\ell, \quad (\text{A.7.2})$$

where C is defined in (32). The initial condition is $\tau_0^* = \sum_{k=1}^K \tau_{0,k}^0 = \sum_{k=1}^K \tau_k^0$. Thus (A.7.1) simplifies to:

$$\tau_\ell^* = \rho \tau_{\ell-1}^* + C \eta^\ell, \quad \tau_0^* \text{ given.} \quad (\text{A.7.3})$$

Step 2: Closed-form solution. Equation (A.7.3) is a first-order linear non-homogeneous difference equation with varying coefficients in the forcing term.

Case 1: $\eta \neq \rho$. Decompose $\tau_\ell^* = \tau_\ell^{(h)} + \tau_\ell^{(p)}$. The homogeneous solution is $\tau_\ell^{(h)} = A \rho^\ell$, and the ansatz $\tau_\ell^{(p)} = B \eta^\ell$ yields:

$$B \eta^\ell = \rho B \eta^{\ell-1} + C \eta^\ell \implies B \eta = \rho B + C \eta \implies B = \frac{C \eta}{\eta - \rho}.$$

Applying the initial condition $\tau_0^* = A + B$:

$$A = \tau_0^* - \frac{C\eta}{\eta - \rho}. \quad (\text{A.7.4})$$

Therefore, for all $L \geq 0$:

$$\tau_L^* = \rho^L \tau_0^* + \frac{C\eta}{\eta - \rho} (\eta^L - \rho^L), \quad (\text{A.7.5})$$

which coincides with equation (34) in the main text.

Case 2: $\eta = \rho$. The recurrence becomes $\tau_\ell^* = \eta \tau_{\ell-1}^* + C\eta^\ell$. Dividing both sides by η^ℓ :

$$\frac{\tau_\ell^*}{\eta^\ell} = \frac{\tau_{\ell-1}^*}{\eta^{\ell-1}} + C. \quad (\text{A.7.6})$$

Defining $u_\ell \equiv \tau_\ell^*/\eta^\ell$, this becomes $u_\ell = u_{\ell-1} + C$, which telescopes to $u_\ell = u_0 + C\ell = \tau_0^* + C\ell$. Restoring τ_L^* :

$$\tau_L^* = \eta^L (\tau_0^* + CL). \quad (\text{A.7.7})$$

Step 3: Vanishing precision (Part (i)). We establish that $\lim_{L \rightarrow \infty} \tau_L^* = 0$ in all cases.

For $\eta \neq \rho$, since $\rho, \eta \in (0, 1)$:

$$\lim_{L \rightarrow \infty} \rho^L = 0, \quad \lim_{L \rightarrow \infty} \eta^L = 0, \quad (\text{A.7.8})$$

so both terms in (A.7.5) converge to zero. For $\eta = \rho$, the linear growth of $\tau_0^* + CL$ is asymptotically dominated by the exponential decay of η^L :

$$\lim_{L \rightarrow \infty} \eta^L (\tau_0^* + CL) = 0, \quad (\text{A.7.9})$$

since $\lim_{L \rightarrow \infty} L^m \eta^L = 0$ for any $m \geq 0$ and $\eta \in (0, 1)$. This completes Part (i).

Step 4: Eventual monotonic decline (Part (ii)). From (A.7.3), the increment is:

$$\Delta_\ell \equiv \tau_\ell^* - \tau_{\ell-1}^* = -(1 - \rho)\tau_{\ell-1}^* + C\eta^\ell. \quad (\text{A.7.10})$$

We prove $\Delta_\ell < 0$ for all sufficiently large ℓ by considering three subcases.

Subcase 4a: $\eta > \rho$. From (A.7.5), as $\ell \rightarrow \infty$, the term η^ℓ dominates ρ^ℓ , yielding:

$$\tau_{\ell-1}^* \sim \frac{C\eta^\ell}{\eta - \rho} \quad (\ell \rightarrow \infty). \quad (\text{A.7.11})$$

Substituting into (A.7.10):

$$\Delta_\ell \sim -(1 - \rho) \frac{C\eta^\ell}{\eta - \rho} + C\eta^\ell = C\eta^\ell \left(\frac{\eta - \rho - 1 + \rho}{\eta - \rho} \right) = C\eta^\ell \frac{\eta - 1}{\eta - \rho}. \quad (\text{A.7.12})$$

Since $\eta < 1$, the numerator $\eta - 1 < 0$, while the denominator $\eta - \rho > 0$. Hence $\Delta_\ell \sim [\text{negative constant}] \times \eta^\ell < 0$ for all large ℓ .

Subcase 4b: $\eta < \rho$. Now ρ^ℓ dominates, giving:

$$\tau_{\ell-1}^* \sim \left(\tau_0^* + \frac{C\eta}{\rho - \eta} \right) \rho^{\ell-1} \quad (\ell \rightarrow \infty). \quad (\text{A.7.13})$$

Substituting into (A.7.10):

$$\Delta_\ell \sim -(1-\rho) \left(\tau_0^* + \frac{C\eta}{\rho-\eta} \right) \rho^{\ell-1} + C\eta^\ell. \quad (\text{A.7.14})$$

Since $\rho > \eta$, the negative term decays as $\rho^{\ell-1}$ while the positive term decays as η^ℓ . The former decays strictly slower and has a negative coefficient, so it dominates for large ℓ . Thus $\Delta_\ell < 0$ for all $\ell \geq \bar{L}$ with:

$$\bar{L} = \inf \left\{ \ell \in \mathbb{N} : \ell > \frac{\ln \left(\frac{(1-\rho)}{C} \left(\tau_0^* + \frac{C\eta}{\rho-\eta} \right) \right)}{\ln(\eta/\rho)} \right\} + 1, \quad (\text{A.7.15})$$

which is finite since $\ln(\eta/\rho) < 0$.

Subcase 4c: $\eta = \rho$. Using (A.7.7):

$$\begin{aligned} \Delta_\ell &= \eta^\ell (\tau_0^* + C\ell) - \eta^{\ell-1} (\tau_0^* + C(\ell-1)) \\ &= \eta^{\ell-1} \left[\underbrace{(\eta-1)\tau_0^*}_{<0} + \underbrace{C(1-(1-\eta)\ell)}_{\rightarrow -\infty} \right]. \end{aligned} \quad (\text{A.7.16})$$

Since the bracket is affine in ℓ with negative slope $-C(1-\eta) < 0$, there exists:

$$\bar{L} = \left\lfloor \frac{\tau_0^*}{C} + \frac{1}{1-\eta} \right\rfloor + 2, \quad (\text{A.7.17})$$

such that $\Delta_\ell < 0$ for all $\ell \geq \bar{L}$. This completes Part (ii).

Step 5: Finite optimal depth (Part (iii)). Strict monotonicity of $V(\cdot)$ and Part (ii) together imply:

$$V(\tau_\ell^*) < V(\tau_{\ell-1}^*) \quad \text{for all } \ell \geq \bar{L}. \quad (\text{A.7.18})$$

Thus $\{V(\tau_\ell^*)\}_{\ell=0}^\infty$ is strictly decreasing beyond $\bar{L}$, so its maximum is attained at some $L^* \in \{0, 1, \dots, \bar{L}\}$.

When the marginal attention cost is $\kappa \geq 0$, adding any layer $\ell \geq \bar{L}$ yields net marginal benefit:

$$\underbrace{V(\tau_\ell^*) - V(\tau_{\ell-1}^*)}_{<0} - \kappa < 0, \quad (\text{A.7.19})$$

which is strictly negative. Hence no planner would ever choose $\ell > \bar{L}$, and $L^* \leq \bar{L} < \infty$ regardless of how small κ is. This completes the proof of Proposition 8. $\blacksquare$

Remark 9 (Comparative statics on $\bar{L}$). *The threshold $\bar{L}$ beyond which precision decreases is increasing in the baseline precision τ_0^* and decreasing in the signal contribution C and the attenuation factor η . Specifically, a higher η (better signal preservation) pushes $\bar{L}$ outward, allowing more layers before returns turn negative. In the limit $\eta \rightarrow 1$, the threshold diverges, recovering the original model where infinite depth can be optimal.*

Remark 10 (Comparison to the no-attenuation case). *When $\eta = 1$ (no signal degradation), the recurrence (A.7.3) becomes $\tau_\ell^* = \rho\tau_{\ell-1}^* + C$, with solution $\tau_L^* = \rho^L\tau_0^* + C(1-\rho^L)/(1-\rho) \rightarrow C/(1-\rho) > 0$. The precision converges to a positive limit, so $V(\tau_L^*)$ converges to $V(C/(1-\rho)) > V(0)$ from below, and infinite depth is optimal whenever $\kappa < V(C/(1-\rho)) - \lim_{L \rightarrow \infty} V(\tau_{L-1}^*) = 0^+$. The diminishing signal quality assumption ($\eta < 1$) is therefore necessary and sufficient for the finite-depth result under uniform technology.*

References

References

- Acemoglu, Daron, and Pascual Restrepo. 2018. “The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment.” *American Economic Review*, 108(6): 1488–1542.
- Acemoglu, Daron, and Pascual Restrepo. 2022. “Tasks, Automation, and the Rise in US Wage Inequality.” *Econometrica*, 90(4): 1973–2016.
- Acemoglu, Daron, David Autor, Jonathon Hazell, and Pascual Restrepo. 2022. “Artificial Intelligence and Jobs: Evidence from Online Vacancies.” *Journal of Labor Economics*, 40(S1): S293–S340.
- Agrawal, Ajay, Joshua Gans, and Avi Goldfarb. 2018. “Prediction, Judgment, and Complexity.” NBER Working Paper No. 24253.
- Bolton, Patrick, and Mathias Dewatripont. 1994. “The Firm as a Communication Network.” *Quarterly Journal of Economics*, 109(4): 809–839.
- Brynjolfsson, Erik, and Lorin Hitt. 2000. “Beyond Computation: Information Technology, Organizational Transformation and Business Performance.” *Journal of Economic Perspectives*, 14(4): 23–48.
- Caplin, Andrew, and Mark Dean. 2015. “Revealed Preference, Rational Inattention, and Costly Information Acquisition.” *American Economic Review*, 105(7): 2183–2203.
- Dessein, Wouter, and Tano Santos. 2006. “Adaptive Organizations.” *Journal of Political Economy*, 114(5): 956–995.
- Dessein, Wouter, Andrea Galeotti, and Tano Santos. 2016. “Rational Inattention and Organizational Focus.” *American Economic Review*, 106(6): 1522–1536.
- Ely, Jeffrey C. 2017. “Beeps.” *American Economic Review*, 107(1): 31–53.
- Escudé, Matteo, and Ludvig Sinander. 2023. “Slow Persuasion.” *Theoretical Economics*, 18(1): 129–162.
- Garicano, Luis. 2000. “Hierarchies and the Organization of Knowledge in Production.” *Journal of Political Economy*, 108(5): 874–904.
- Hadfield-Menell, Dylan. 2021. “The Principal-Agent Alignment Problem in Artificial Intelligence.” PhD diss., University of California, Berkeley.
- Holmström, Bengt, and Paul Milgrom. 1991. “Multitask Principal-Agent Analyses.” *Journal of Law, Economics, & Organization*, 7: 24–52.
- Ide, Enrique, and Eduard Talamàs. 2025. “Artificial Intelligence in the Knowledge Economy.” *Journal of Political Economy*, 133(2): 3762–3800.
- Kamenica, Emir, and Matthew Gentzkow. 2011. “Bayesian Persuasion.” *American Economic Review*, 101(6): 2590–2615.

- Klein, Nicolas, and Thomas Wiecezorek. 2026. “Artificial Intelligence and Organizational Structure.” Forthcoming, *Quarterly Journal of Economics*, 141(1).
- Lash, Alex, and David Wellington. 2008. “Information Processing in Hierarchical Organizations.” *Management Science*, 54(8): 1423–1439.
- Lien, Jaimie W., Xincheng Qiu, and Xiaolan Zhong. 2019. “Hierarchies and Information Processing in Teams.” *RAND Journal of Economics*, 50(3): 671–696.
- Mackowiak, Bartosz, Filip Matějka, and Mirko Wiederholt. 2023. “Rational Inattention: A Review.” *Journal of Economic Literature*, 61(1): 226–273.
- Melumad, Nahum D., Dilip Mookherjee, and Stefan Reichelstein. 1995. “Hierarchical Decentralization of Incentive Contracts.” *RAND Journal of Economics*, 26(4): 654–672.
- Noy, Shakked, and Whitney Zhang. 2023. “Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence.” *Science*, 381(6654): 187–192.
- Peng, Sida, Eirini Kalliamvakou, Peter Cihon, and Mert Demirer. 2023. “The Impact of AI on Developer Productivity: Evidence from GitHub Copilot.” arXiv preprint arXiv:2302.06590.
- Phelps, Steve, and Tom Ranson. 2023. “Of Models and Tin Men: A Multi-Task Principal-Agent Analysis of Large Language Models.” *Journal of Artificial Intelligence Research*, 77: 1123–1156.
- Radner, Roy. 1993. “The Organization of Decentralized Information Processing.” *Econometrica*, 61(5): 1109–1146.
- Renault, Jérôme, Eilon Solan, and Nicolas Vieille. 2017. “Optimal Dynamic Information Provision.” *Games and Economic Behavior*, 104: 329–349.
- Sims, Christopher A. 2003. “Implications of Rational Inattention.” *Journal of Monetary Economics*, 50(3): 665–690.
- Sims, Christopher A. 2006. “Rational Inattention: Beyond the Linear-Quadratic Case.” *American Economic Review*, 96(2): 158–163.
- Smolin, Alex. 2021. “Dynamic Evaluation Design.” *American Economic Journal: Microeconomics*, 13(4): 300–331.
- Tomašev, Nenad, Ulrich Paquet, and Karl Tuyls. 2026. “Intelligent AI Delegation: A Survey and Framework.” arXiv:2602.11865.
- Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. “Attention Is All You Need.” *Advances in Neural Information Processing Systems*, 30: 5998–6008.
- Williamson, Oliver E. 1985. *The Economic Institutions of Capitalism*. New York: Free Press.
- Yang, Zonghao, and Liwen Zhai. 2025. “Ten Principles of AI Agent Economics.” arXiv:2505.20273.